



## *Analytical geometry*

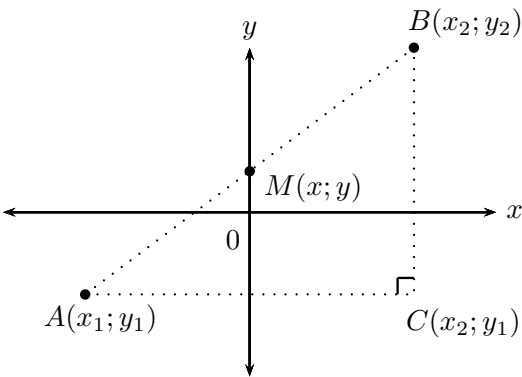
|     |                                          |     |
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7.1 Revision

EMCHN

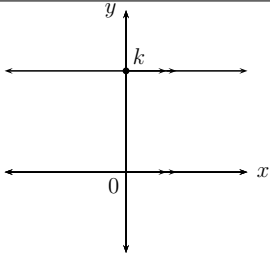
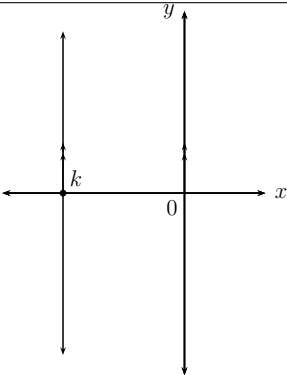
Straight line equations

EMCHP



|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| Theorem of Pythagoras:       | $AB^2 = AC^2 + BC^2$                                                             |
| Distance formula:            | $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$                                      |
| Gradient:                    | $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$ or $m_{AB} = \frac{y_1 - y_2}{x_1 - x_2}$ |
| Mid-point of a line segment: | $M(x; y) = \left( \frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$              |
| Points on a straight line:   | $m_{AB} = m_{AM} = m_{MB}$                                                       |

|                          |                                                         |  |
|--------------------------|---------------------------------------------------------|--|
| Two-point form:          | $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ |  |
| Gradient-point form:     | $y - y_1 = m(x - x_1)$                                  |  |
| Gradient-intercept form: | $y = mx + c$                                            |  |

|                   |         |                                                                                    |
|-------------------|---------|------------------------------------------------------------------------------------|
| Horizontal lines: | $y = k$ |  |
| Vertical lines:   | $x = k$ |  |

### Worked example 1: Revision

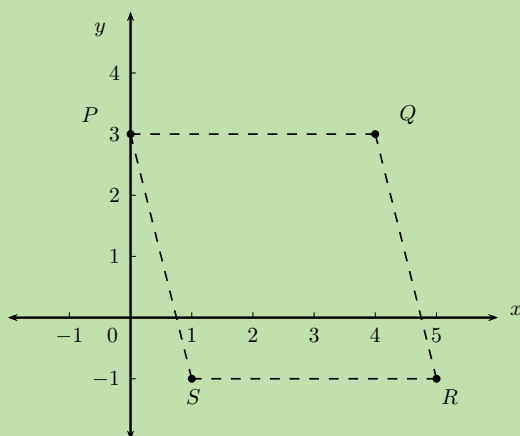
#### QUESTION

Given quadrilateral  $PQRS$  with vertices  $P(0; 3)$ ,  $Q(4; 3)$ ,  $R(5; -1)$  and  $S(1; -1)$ .

1. Determine the equation of the lines  $PS$  and  $QR$ .
2. Show that  $PS \parallel QR$ .
3. Calculate the lengths of  $PS$  and  $QR$ .
4. Determine the equation of the diagonal  $QS$ .
5. What type of quadrilateral is  $PQRS$ ?

#### SOLUTION

##### Step 1: Draw a sketch



**Step 2: Use the given information to determine the equation of lines  $PS$  and  $QR$**

$$\text{Gradient: } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Two-point form: } \frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Gradient-intercept form: } y = mx + c$$

Determine the equation of the line  $PS$  using the two point form of the straight line equation:

$$x_1 = 0; \quad y_1 = 3; \quad x_2 = 1; \quad y_2 = -1$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - 3}{x - 0} = \frac{-1 - 3}{1 - 0}$$

$$\frac{y - 3}{x} = -4$$

$$y - 3 = -4x$$

$$\therefore y = -4x + 3$$

Determine the equation of the line  $QR$  using the gradient-intercept form of the straight line equation:

$$m_{QR} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1 - 3}{5 - 4}$$

$$= \frac{-4}{1}$$

$$= -4$$

$$y = mx + c$$

$$y = -4x + c$$

$$\text{Substitute } (4; 3) \quad 3 = -4(4) + c$$

$$\therefore c = 19$$

$$y = -4x + 19$$

There is often more than one method for determining the equation of a line. The different forms of the straight line equation are used, depending on the information provided in the problem.

**Step 3: Show that line  $PS$  and line  $QR$  have equal gradients**

$$y = -4x + 3$$

$$\therefore m_{PS} = -4$$

$$\text{And } y = -4x + 19$$

$$\therefore m_{QR} = -4$$

$$\therefore m_{PS} = m_{QR}$$

$$\therefore PS \parallel QR$$

**Step 4: Use the distance formula to determine the lengths of  $PS$  and  $QR$**

$$\begin{aligned} PS &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(1 - 0)^2 + (-1 - 3)^2} \\ &= \sqrt{1 + (-4)^2} \\ &= \sqrt{17} \text{ units} \end{aligned}$$

$$\begin{aligned} QR &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(5 - 4)^2 + (-1 - 3)^2} \\ &= \sqrt{1 + (-4)^2} \\ &= \sqrt{17} \text{ units} \end{aligned}$$

**Step 5: Determine the equation of the diagonal  $QS$**

Determine the gradient of the line:

$$\begin{aligned} m_{QS} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-1 - 3}{1 - 4} \\ &= \frac{-4}{-3} \\ &= \frac{4}{3} \end{aligned}$$

Use gradient and the point  $Q(4; 3)$  to determine the equation of the line  $QS$ :

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - y_1 &= \frac{4}{3}(x - x_1) \\ \text{Substitute } (4; 3) \quad y - 3 &= \frac{4}{3}(x - 4) \\ y - 3 &= \frac{4}{3}x - \frac{16}{3} \\ y &= \frac{4}{3}x - \frac{16}{3} + 3 \\ \therefore y &= \frac{4}{3}x - \frac{7}{3} \end{aligned}$$

**Step 6: Examine the properties of quadrilateral  $PQRS$**

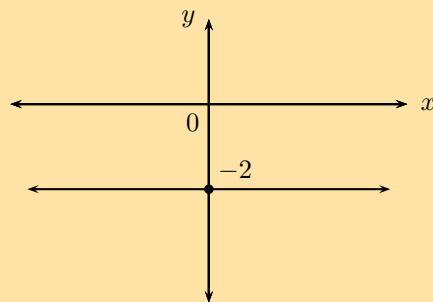
We have shown that  $PS \parallel QR$  and  $PS = QR$ , therefore quadrilateral  $PQRS$  is a parallelogram (one pair of opposite sides equal and parallel).

### Exercise 7 – 1: Revision

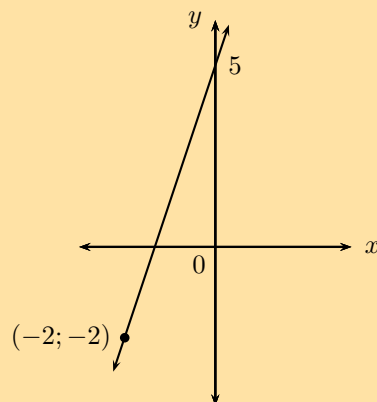
1. Determine the following for the line segment between the given points:

- |                             |                                     |
|-----------------------------|-------------------------------------|
| • length                    | • gradient                          |
| • mid-point                 | • equation                          |
| a) $(-2; -4)$ and $(3; 11)$ | c) $(h; -h - k)$ and $(2k; h - 5k)$ |
| b) $(-5; -3)$ and $(10; 6)$ | d) $(2; 9)$ and $(0; -1)$           |

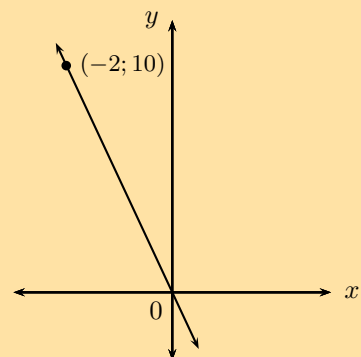
2. The line joining  $A(x; y)$  and  $B(-3; 6)$  has the mid-point  $M(2; 3)$ . Determine the values of  $x$  and  $y$ .
3. Given  $F(2; 11)$ ,  $G(-4; r)$  and length  $FG = 6\sqrt{5}$  units, determine the value(s) of  $r$ .
4. Determine the equation of the straight line:
  - a) passing through the point  $(\frac{1}{2}; 4)$  and  $(1; 5)$ .
  - b) passing through the points  $(2; -3)$  and  $(-1; 0)$ .
  - c) passing through the point  $(9; 1)$  and with  $m = \frac{1}{3}$ .
  - d) parallel to the  $x$ -axis and passing through the point  $(0; -4)$ .
  - e) passing through the point  $(\frac{1}{2}; -1)$  and with  $m = -4$ .
  - f) perpendicular to the  $x$ -axis and passing through the point  $(5; 0)$ .
  - g) with undefined gradient and passing through the point  $(\frac{3}{4}; 0)$ .
  - h) with  $m = 2p$  and passing through the point  $(3; 6p + 3)$ .
  - i) which cuts the  $y$ -axis at  $y = -\frac{3}{5}$  and with  $m = 4$ .
  - j)



k)



l)



5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

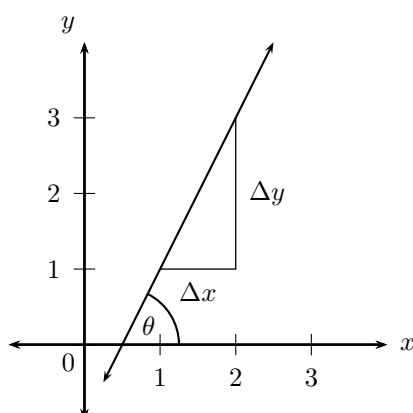
- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 293J | 1b. 293K | 1c. 293M | 1d. 293N | 2. 293P  | 3. 293Q  |
| 4a. 293R | 4b. 293S | 4c. 293T | 4d. 293V | 4e. 293W | 4f. 293X |
| 4g. 293Y | 4h. 293Z | 4i. 2942 | 4j. 2943 | 4k. 2944 | 4l. 2945 |



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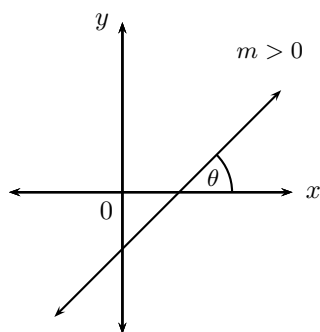
[m.everythingmaths.co.za](http://m.everythingmaths.co.za)



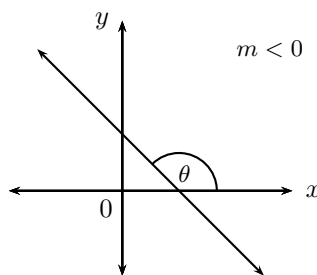
The diagram shows a straight line which forms an acute angle  $\theta$  with the positive  $x$ -axis. This is called the **angle of inclination** of a straight line.

The gradient of a straight line is equal to the tangent of the angle formed between the line and the positive direction of the  $x$ -axis.

$$m = \tan \theta \quad \text{for } 0^\circ \leq \theta < 180^\circ$$



$\theta$  is an acute angle



$\theta$  is an obtuse angle

### Lines with positive gradients

A line with a positive gradient ( $m > 0$ ) has an acute angle of inclination ( $0^\circ < \theta < 90^\circ$ ).

For example, we can determine the angle of inclination of a line with  $m = 1,2$ :

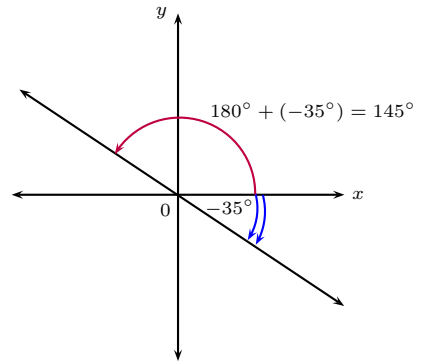
$$\begin{aligned} \tan \theta &= m \\ &= 1,2 \\ \therefore \theta &= \tan^{-1}(1,2) \\ &= 50,2^\circ \end{aligned}$$

## Lines with negative gradients

If we are calculating the angle of inclination of a line with a negative gradient ( $m < 0$ ), then we add  $180^\circ$  to change the negative angle to an obtuse angle ( $90^\circ < \theta < 180^\circ$ ).

For example, we can determine the angle of inclination for a line with  $m = -0,7$ :

$$\begin{aligned}\tan \theta &= m \\ &= -0,7 \\ \therefore \theta &= \tan^{-1}(-0,7) \\ &= -35,0^\circ \\ \text{Obtuse angle: } \theta &= -35,0^\circ + 180^\circ \\ &= 145^\circ\end{aligned}$$



### Worked example 2: Inclination of a straight line

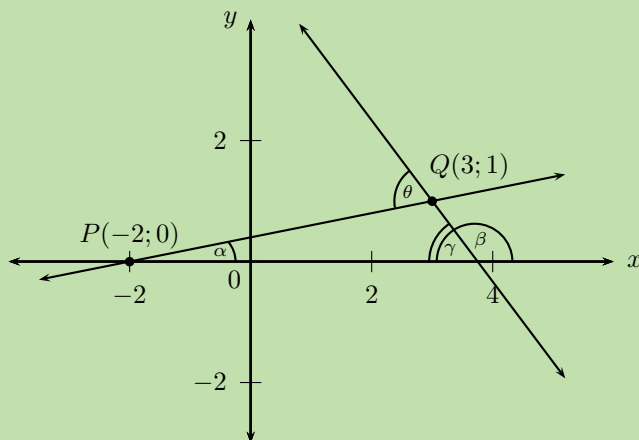
#### QUESTION

Determine the acute angle (correct to 1 decimal place) between the line passing through the points  $P(-2; 0)$  and  $Q(3; 1)$  and the straight line  $y = -\frac{4}{3}x + 5$ .

#### SOLUTION

##### Step 1: Draw a sketch

Draw the line through points  $P(-2; 0)$  and  $Q(3; 1)$  and the line  $y = -\frac{4}{3}x + 5$  on a suitable system of axes. Label  $\alpha$  and  $\beta$ , the angles of inclination of the two lines. Label  $\theta$ , the acute angle between the two straight lines.





Notice that  $\alpha$  and  $\theta$  are acute angles and  $\beta$  is an obtuse angle.

$$\gamma = 180^\circ - \beta \quad (\angle \text{ on str. line})$$

$$\text{and } \theta = \alpha + \gamma \quad (\text{ext. } \angle \text{ of } \triangle = \text{sum int. opp})$$

$$\therefore \theta = \alpha + (180^\circ - \beta)$$

$$= 180^\circ + \alpha - \beta$$

### Step 2: Use the gradient to determine the angle of inclination $\beta$

From the equation  $y = -\frac{4}{3}x + 5$  we see that  $m < 0$ , therefore  $\beta$  is an obtuse angle.

$$m = -\frac{4}{3}$$

$$\tan \beta = -\frac{4}{3}$$

$$\therefore \beta = \tan^{-1} \left( -\frac{4}{3} \right)$$

$$= -53,1^\circ$$

$$\beta = -53,1^\circ + 180^\circ$$

$$= 126,9^\circ$$

### Step 3: Determine the gradient and angle of inclination of the line through $P$ and $Q$

Determine the gradient

$$m = \frac{y_P - y_Q}{x_P - x_Q}$$

$$= \frac{-1}{-5}$$

$$= \frac{1}{5}$$

Determine the angle of inclination

$$\tan \alpha = m$$

$$= \frac{1}{5}$$

$$\therefore \alpha = \tan^{-1} \left( \frac{1}{5} \right)$$

$$= 11,3^\circ$$

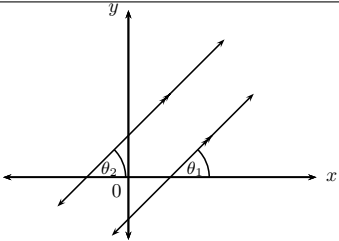
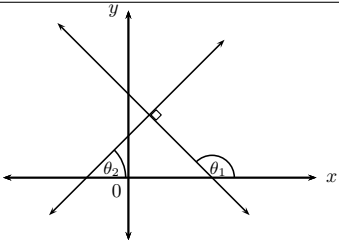
### Step 4: Write the final answer

$$\theta = 180^\circ + \alpha - \beta$$

$$= 180^\circ + 11,3^\circ - 126,9^\circ$$

$$= 64,4^\circ$$

The acute angle between the two straight lines is  $64,4^\circ$ .

|                     |                                                                                   |                       |                                  |
|---------------------|-----------------------------------------------------------------------------------|-----------------------|----------------------------------|
| Parallel lines      |  | $m_1 = m_2$           | $\theta_1 = \theta_2$            |
| Perpendicular lines |  | $m_1 \times m_2 = -1$ | $\theta_1 = 90^\circ + \theta_2$ |

### Worked example 3: Parallel lines

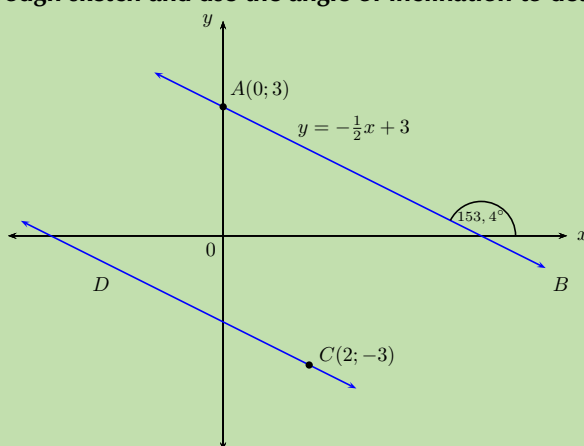
#### QUESTION

Line  $AB$  passes through the point  $A(0; 3)$  and has an angle of inclination of  $153,4^\circ$ .

- Determine the equation of line  $CD$  which passes through the point  $C(2; -3)$  and is parallel to  $AB$ .
- Determine the equation of line  $EF$ , which passes through the origin and is perpendicular to both  $AB$  and  $CD$ .
- Sketch lines  $AB$ ,  $CD$  and  $EF$  on the same system of axes.
- Use two different methods to determine the angle of inclination of  $EF$ .

#### SOLUTION

**Step 1: Draw a rough sketch and use the angle of inclination to determine the equation of  $CD$**



$$\begin{aligned}
 m_{AB} &= \tan \theta \\
 &= \tan 153,4^\circ \\
 &= -0,5
 \end{aligned}$$

Since we are given  $AB \parallel CD$ ,

$$m_{CD} = m_{AB} = -0,5 = -\frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = -\frac{1}{2}(x - x_1)$$

Substitute the given point  $(2; -3)$ :

$$y - (-3) = -\frac{1}{2}(x - 2)$$

$$y + 3 = -\frac{1}{2}x + 1$$

$$y = -\frac{1}{2}x - 2$$

### Step 2: Determine the equation of $EF$

$EF$  is perpendicular to  $AB$ , therefore the product of their gradients is equal to  $-1$ :

$$m_{AB} \times m_{EF} = -1$$

$$-\frac{1}{2} \times m_{EF} = -1$$

$$\therefore m_{EF} = 2$$

We know line  $EF$  passes through  $(0; 0)$ , therefore the equation of the line is:

$$y = 2x$$

### Step 3: Determine the angle of inclination of $EF$

Let the angle of inclination of  $EF$  be  $\beta$ .

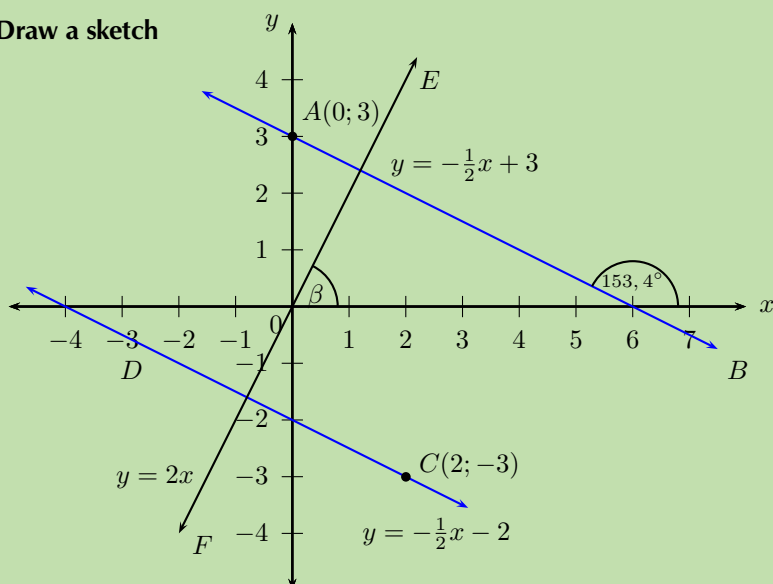
Method 1:

$$\begin{aligned}\beta &= 153,4^\circ - 90^\circ \\ &= 63,4^\circ\end{aligned}$$

Method 2:

$$\begin{aligned}m &= 2 \\ \tan \beta &= 2 \\ \therefore \beta &= 63,4^\circ\end{aligned}$$

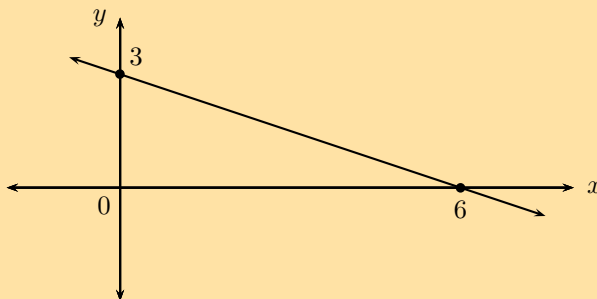
### Step 4: Draw a sketch



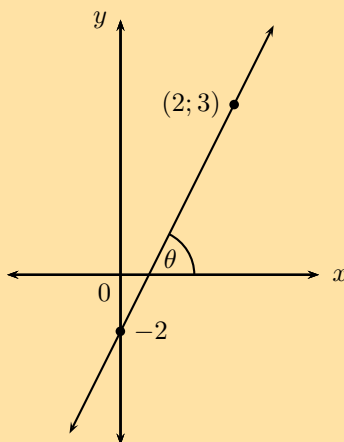
## Exercise 7 – 2: Inclination of a straight line

1. Determine the angle of inclination (correct to 1 decimal place) for each of the following:

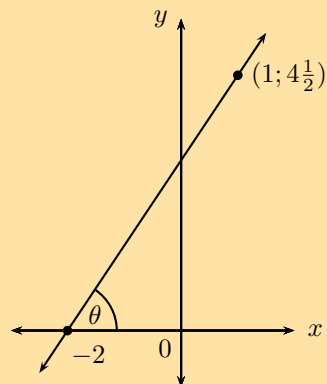
- a) a line with  $m = \frac{3}{4}$
- b)  $6 + x = 2y$
- c) the line passes through the points  $(-4; 0)$  and  $(2; 6)$
- d)  $y = 4$
- e) a line with a gradient of 1,733
- f)



g)



h)



- 2. Find the angle between the line  $2y = 5x$  and the line passing through points  $T(2; 1\frac{1}{3})$  and  $V(-3; 3)$ .
- 3. Determine the equation of the straight line that passes through the point  $(1; 2)$  and is parallel to the line  $y + 3x = 1$ .
- 4. Determine the equation of the straight line that passes through the point  $(-4; -4)$  and is parallel to the line with angle of inclination  $\theta = 56,31^\circ$ .
- 5. Determine the equation of the straight line that passes through the point  $(1; -6)$  and is perpendicular to the line  $5y = x$ .
- 6. Determine the equation of the straight line that passes through the point  $(3; -1)$  and is perpendicular to the line with angle of inclination  $\theta = 135^\circ$ .
- 7.  $A(2; 3)$ ,  $B(-4; 0)$  and  $C(5; -3)$  are the vertices of  $\triangle ABC$  in the Cartesian plane.  $AC$  intersects the  $x$ -axis at  $D$ . Draw a sketch and determine the following:
  - a) the equation of line  $AC$
  - b) the coordinates of point  $D$

- c) the angle of inclination of  $AC$
- d) the gradient of line  $AB$
- e)  $\hat{BAC}$
- f) the equation of the line perpendicular to  $AB$  and passing through the origin
- g) the mid-point  $M$  of  $BC$
- h) the equation of the line parallel to  $AC$  and passing through point  $M$

8. Points  $F(-3; 5)$ ,  $G(-7; -4)$  and  $H(2; 0)$  are given.

- a) Plot the points on the Cartesian plane.
- b) Determine the coordinates of  $I$  if  $FGHI$  is a parallelogram.
- c) Prove that  $FGHI$  is a rhombus.

9. Given points  $S(2; 5)$ ,  $T(-3; -4)$  and  $V(4; -2)$ .

- a) Show that the equation of the line  $ST$  is  $5y = 9x + 7$ .
- b) Determine the size of  $\hat{TSV}$ .

10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|
| 1a. 2946 | 1b. 2947 | 1c. 2948 | 1d. 2949 | 1e. 294B | 1f. 294C |
| 1g. 294D | 1h. 294F | 2. 294G  | 3. 294H  | 4. 294J  | 5. 294K  |
| 6. 294M  | 7. 294N  | 8. 294P  | 9. 294Q  |          |          |



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## 7.2 Equation of a circle

EMCHS

### Equation of a circle with centre at the origin

EMCHT

#### Investigation:

1. Draw a system of axes with a scale of  $1 \text{ cm} = 1 \text{ unit}$  on the  $x$ -axis and on the  $y$ -axis.
2. Draw the lines  $y = x$  and  $y = -x$ .
3. Plot the following points:
 

|                             |                              |
|-----------------------------|------------------------------|
| a) $O(0; 0)$                | f) $H(-2; 0)$                |
| b) $D(2; 0)$                | g) $I(-\sqrt{2}; -\sqrt{2})$ |
| c) $E(\sqrt{2}; \sqrt{2})$  | h) $J(0; -2)$                |
| d) $F(0; 2)$                | i) $K(\sqrt{2}; -\sqrt{2})$  |
| e) $G(-\sqrt{2}; \sqrt{2})$ |                              |

What object do the points form?

4. Measure the following distances and complete the table below:

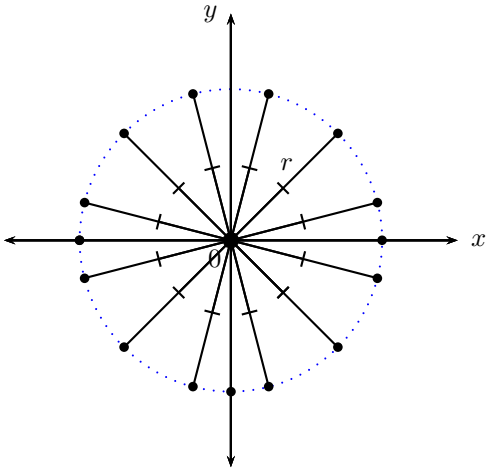
| Line segment | Distance ( <i>cm</i> ) |
|--------------|------------------------|
| <i>DO</i>    |                        |
| <i>EO</i>    |                        |
| <i>FO</i>    |                        |
| <i>GO</i>    |                        |
| <i>HO</i>    |                        |
| <i>IO</i>    |                        |
| <i>JO</i>    |                        |
| <i>KO</i>    |                        |

5. Use the distance formula to check the results of the above table:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

| Line segment | Distance |
|--------------|----------|
| <i>DO</i>    |          |
| <i>EO</i>    |          |
| <i>FO</i>    |          |
| <i>GO</i>    |          |
| <i>HO</i>    |          |
| <i>IO</i>    |          |
| <i>JO</i>    |          |
| <i>KO</i>    |          |

- What do you notice about the length of each line segment?
- What is the general term given to this type of line segment?
- If the point  $P(x; y)$  lies on the circle, use the distance formula to determine an expression for the length of  $PO$ .
- Can you deduce a general equation for a circle with centre at the origin?

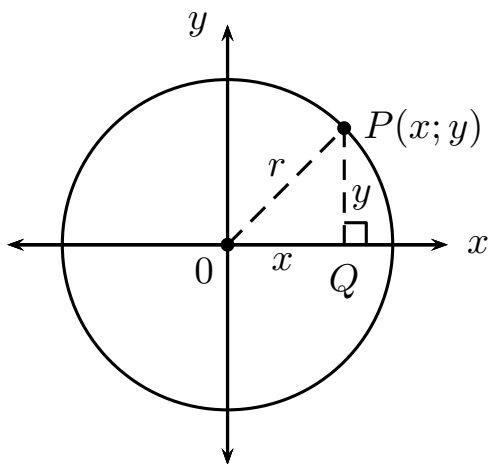


A circle is the set of all points that are an equal distance (radius) from a given point (centre). In other words, every point on the circumference of a circle is equidistant from its centre.

The radius of a circle is the distance from the centre of a circle to any point on the circumference.

A diameter of a circle is any line passing through the centre of the circle which connects two points on the circle. The diameter is also the name given to the maximum distance between two points on a circle.

Consider a point  $P(x; y)$  on the circumference of a circle of radius  $r$  with centre at  $O(0; 0)$ .



In  $\triangle OPQ$ :

$$OP^2 = PQ^2 + OQ^2 \quad (\text{Pythagoras})$$

$$OP = r$$

$$PQ = y - 0$$

$$OQ = x - 0$$

$$r^2 = (y - 0)^2 + (x - 0)^2$$

$$\therefore r^2 = x^2 + y^2$$

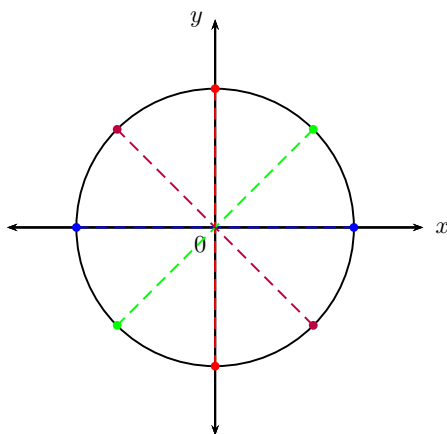
#### Equation of a circle with centre at the origin:

If  $P(x; y)$  is a point on a circle with centre  $O(0; 0)$  and radius  $r$ , then the equation of the circle is:

$$x^2 + y^2 = r^2$$

#### Circle symmetry

A circle with centre  $(0; 0)$  is symmetrical about the origin: for every point  $(x; y)$  on the circumference of a circle, there is also the point  $(-x; -y)$ .



A circle centred on the origin is also symmetrical about the  $x$ - and  $y$ -axis. Is a circle centred on the origin symmetrical about the lines  $y = x$  and  $y = -x$ ? How many lines of symmetry does a circle have?

#### Worked example 4: Equation of a circle with centre at the origin

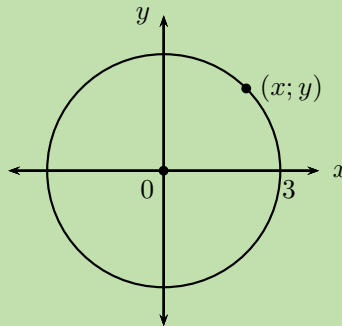
##### QUESTION

Given: circle with centre  $O(0; 0)$  and a radius of 3 units.

1. Sketch the circle on the Cartesian plane.
2. Determine the equation of the circle.
3. Show that the point  $T(-\sqrt{4}; \sqrt{5})$  lies on the circle.

##### SOLUTION

###### Step 1: Draw a sketch



###### Step 2: Determine the equation of the circle

Write down the general form of the equation of a circle with centre  $(0; 0)$ :

$$x^2 + y^2 = r^2$$

Substitute  $r = 3$  :  $x^2 + y^2 = (3)^2$

$$x^2 + y^2 = 9$$

###### Step 3: Show that point $T$ lies on the circle

Substitute the  $x$ -coordinate and the  $y$ -coordinate into the left-hand side of the equation and show that it is equal to the right-hand side:

$$\begin{aligned}\text{LHS} &= x^2 + y^2 \\ &= (-\sqrt{4})^2 + (\sqrt{5})^2 \\ &= 4 + 5 \\ &= 9 \\ &= r^2 \\ &= \text{RHS}\end{aligned}$$

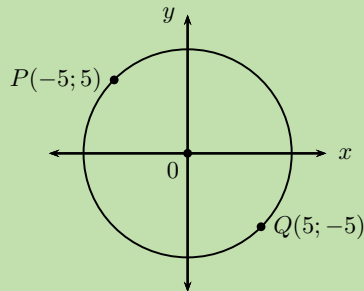
Therefore,  $T(-\sqrt{4}; \sqrt{5})$  lies on the circle  $x^2 + y^2 = 9$ .



**QUESTION**

A circle with centre  $O(0; 0)$  passes through the points  $P(-5; 5)$  and  $Q(5; -5)$ .

1. Plot the points and draw a rough sketch of the circle.
2. Determine the equation of the circle.
3. Calculate the length of  $PQ$ .
4. Explain why  $PQ$  is a diameter of the circle.

**SOLUTION****Step 1: Draw a sketch****Step 2: Determine the equation of the circle**

Write down the general form of the equation of a circle with centre  $(0; 0)$  and substitute  $P(-5; 5)$ :

$$\begin{aligned}x^2 + y^2 &= r^2 \\(-5)^2 + (5)^2 &= r^2 \\25 + 25 &= r^2 \\50 &= r^2 \\\therefore r &= \sqrt{50} \quad (r \text{ is always positive}) \\r &= 5\sqrt{2} \text{ units}\end{aligned}$$

Therefore, the equation of the circle passing through  $P$  and  $Q$  is  $x^2 + y^2 = 50$ .

**Step 3: Calculate the length  $PQ$** 

Use the distance formula to determine the distance between the two points.

$$\begin{aligned}PQ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\&= \sqrt{(5 - (-5))^2 + (-5 - 5)^2} \\&= \sqrt{(10)^2 + (-10)^2} \\&= \sqrt{100 + 100} \\&= \sqrt{100 \cdot 2} \\&= 10\sqrt{2} \text{ units}\end{aligned}$$

**Step 4: Determine if  $PQ$  is a diameter of the circle**

$$\begin{aligned}r &= 5\sqrt{2} \\ \text{And } d &= 2 \times r \\ &= 2 \times 5\sqrt{2} \\ &= 10\sqrt{2} \\ \therefore d &= PQ\end{aligned}$$

Since  $PQ$  connects two points,  $P$  and  $Q$ , on the circle and is a line of length  $10\sqrt{2}$  units,  $PQ$  is a diameter of the circle.

**Alternative method 1:** Use symmetry to show that  $PQ$  is a diameter of the circle.

The diameter is the name given to the maximum distance between two points on a circle; this means that the two points must lie opposite each other with respect to the centre.

Using symmetry about the origin, we know that  $(x; y)$  lies opposite  $(-x; -y)$  on the circle and vice versa:

- $P(-5; 5)$  lies opposite  $(5; -5)$ , which are the coordinates of  $Q$
- $Q(5; -5)$  lies opposite  $(-5; 5)$ , which are the coordinates of  $P$

Therefore,  $PQ$  is a diameter of the circle.

**Alternative method 2:** Show that  $PQ$  passes through the centre.

Determine the equation of  $PQ$  and show that it passes through the origin.

$$\begin{aligned}\frac{y - y_Q}{x - x_Q} &= \frac{y_P - y_Q}{x_P - x_Q} \\ \frac{y + 5}{x - 5} &= \frac{5 + 5}{-5 - 5} \\ \frac{y + 5}{x - 5} &= \frac{10}{-10} \\ y + 5 &= -(x - 5) \\ y &= -x + 5 - 5 \\ y &= -x\end{aligned}$$

$\therefore (0; 0)$  lies on the line  $PQ$ .

Therefore,  $PQ$  passes through the centre and is a diameter of the circle.

### Worked example 6: Equation of a circle with centre at the origin

#### QUESTION

Given a circle with centre  $O(0; 0)$  and a radius of  $\sqrt{45}$  units. Determine the possible coordinates of the point(s) on the circle which have an  $x$ -value that is twice the  $y$ -value.

#### SOLUTION

##### Step 1: Determine the equation of the circle

$$\begin{aligned}x^2 + y^2 &= r^2 \\x^2 + y^2 &= (\sqrt{45})^2 \\x^2 + y^2 &= 45\end{aligned}$$

##### Step 2: Determine the coordinates of the points on the circle

To calculate the possible coordinates of the point(s) on the circle which have an  $x$ -value that is twice the  $y$ -value, we substitute  $x = 2y$  into the equation of the circle:

$$\begin{aligned}x^2 + y^2 &= 45 & \text{If } y = -3 : & \quad x = 2(-3) \\(2y)^2 + y^2 &= 45 & & \quad = -6 \\4y^2 + y^2 &= 45 & \text{If } y = 3 : & \quad x = 2(3) \\5y^2 &= 45 & & \quad = 6 \\y^2 &= 9 \\ \therefore y &= \pm 3\end{aligned}$$

This gives the points  $(6; 3)$  and  $(-6; -3)$ .

Note: we can check that both points lie on the circle by substituting the coordinates into the equation of the circle:

$$\begin{aligned}(6)^2 + (3)^2 &= 36 + 9 = 45 \\(-6)^2 + (-3)^2 &= 36 + 9 = 45\end{aligned}$$

### Exercise 7 – 3: Equation of a circle with centre at the origin

1. Complete the following for each circle given below:

- Determine the radius.
- Draw a sketch.
- Calculate the coordinates of two points on the circle.

a)  $x^2 + y^2 = 16$

d)  $y^2 = 20 - x^2$

b)  $x^2 + y^2 = 100$

e)  $x^2 + y^2 = 2,25$

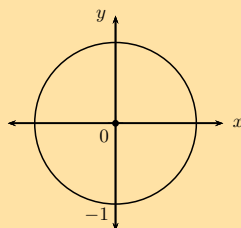
c)  $3x^2 + 3y^2 = 27$

f)  $y^2 = -x^2 + \frac{10}{9}$

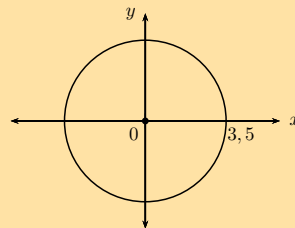
2. Determine the equation of the circle:

- a) with centre at the origin and a radius of 5 units.
- b) with centre at  $(0; 0)$  and  $r = \sqrt{11}$  units.
- c) passing through the point  $(3; 5)$  and with centre  $(0; 0)$ .
- d) centred at the origin and  $r = 2,5$  units.
- e) with centre at the origin and a diameter of 30 units.
- f) passing through the point  $(p; 3q)$  and with centre at the origin.

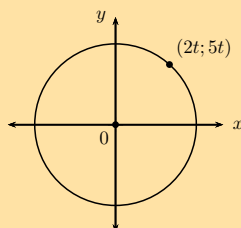
g)



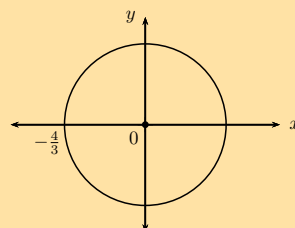
i)



h)



j)



3. Determine whether or not the following equations represent a circle:

- a)  $x^2 + y^2 - 8 = 0$
- b)  $y^2 - x^2 + 25 = 0$
- c)  $3x^2 + 6y^2 = 18$
- d)  $x^2 = \sqrt{6} - y^2$
- e)  $y(y + x) = -x(x - y) + 11$
- f)  $\sqrt{80} + x^2 - y^2 = 0$
- g)  $\frac{y^2}{3} + \frac{x^2}{3} = 3$

4. Determine the value(s) of  $g$  if  $(\sqrt{3}; g)$  is a point on the circle  $x^2 + y^2 = 19$ .

5.  $A(s; t)$  is a point on the circle with centre at the origin and a diameter of 40 cm.

- a) Determine the possible coordinates of  $A$  if the value of  $s$  is triple the value of  $t$ .
- b) Determine the possible coordinates of  $A$  if the value of  $s$  is half the value of  $t$ .

6.  $P(-2; 3)$  lies on a circle with centre at  $(0; 0)$ .

- a) Determine the equation of the circle.
- b) Sketch the circle and label point  $P$ .
- c) If  $PQ$  is a diameter of the circle, determine the coordinates of  $Q$ .
- d) Calculate the length of  $PQ$ .
- e) Determine the equation of the line  $PQ$ .
- f) Determine the equation of the line perpendicular to  $PQ$  and passing through the point  $P$ .

7. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 294S    1b. 294T    1c. 294V    1d. 294W    1e. 294X    1f. 294Y  
 2a. 294Z    2b. 2952    2c. 2953    2d. 2954    2e. 2955    2f. 2956  
 2g. 2957    2h. 2958    2i. 2959    2j. 295B    3a. 295C    3b. 295D  
 3c. 295F    3d. 295G    3e. 295H    3f. 295J    3g. 295K    4. 295M  
 5. 295N    6. 295P



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## Equation of a circle with centre at $(a; b)$

EMCHV

### Investigation: Shifting the centre of a circle

Complete the following for each equation in the table below:

- Write down the shifted equation (do not simplify).
- Draw a rough sketch to illustrate the shift(s).

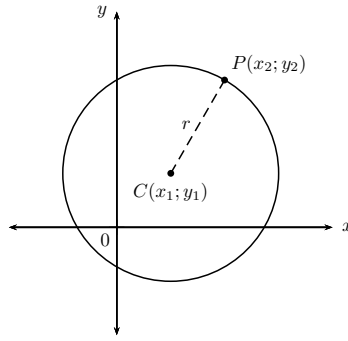
1. Vertical shift: the graph is shifted 1 unit up.
2. Horizontal shift: the graph is shifted 2 units to the right.
3. Combined shifts: the graph is shifted 1 unit up and 2 units to the right.

The first example has been completed.

| Equation        | Vertical shift       | Horizontal shift         | Combined shift                 |
|-----------------|----------------------|--------------------------|--------------------------------|
| $y - 3x^2 = 0$  | $(y - 1) - 3x^2 = 0$ | $y - 3(x - 2)^2 = 0$     | $(y - 1) - 3(x - 2)^2 = 0$     |
| $y = 3x^2$<br>  | $y - 1 = 3x^2$<br>   | $y - 3(x - 2)^2 = 0$<br> | $(y - 1) - 3(x - 2)^2 = 0$<br> |
| $y - 5^x = 0$   |                      |                          |                                |
|                 |                      |                          |                                |
| $x^2 + y^2 = 4$ |                      |                          |                                |
|                 |                      |                          |                                |

Use the table to answer the following questions:

1. Write down the general equation of a circle with centre  $(0; 0)$ .
2. Write down the general equation of a circle with centre  $(0; b)$ .
3. Write down the general equation of a circle with centre  $(a; 0)$ .
4. Write down the general equation of a circle with centre  $(a; b)$ .



Consider a circle in the Cartesian plane with centre at  $C(x_1; y_1)$  and with a radius of  $r$  units. If  $P(x_2; y_2)$  is any point on the circumference of the circle, we can use the distance formula to calculate the distance between the two points:

$$PC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

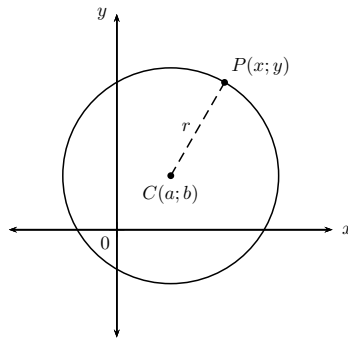
The distance  $PC$  is equal to the radius ( $r$ ) of the circle:

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\therefore r^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

If the coordinates of the centre of the circle are  $(a; b)$ , then the equation of a circle not centred on the origin is:

$$(x_2 - a)^2 + (y_2 - b)^2 = r^2$$



#### Equation of a circle with centre at $(a; b)$ :

If  $P(x; y)$  is a point on a circle with centre  $C(a; b)$  and radius  $r$ , then the equation of the circle is:

$$(x - a)^2 + (y - b)^2 = r^2$$

A circle with centre  $(0; 0)$  is a special case of the general equation:

$$(x - a)^2 + (y - b)^2 = r^2$$

$$(x - 0)^2 + (y - 0)^2 = r^2$$

$$\therefore x^2 + y^2 = r^2$$

### Worked example 7: Equation of a circle with centre at $(a; b)$

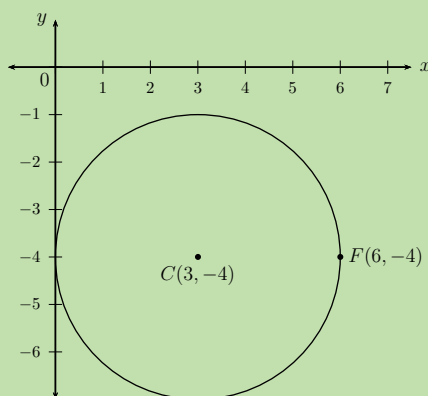
#### QUESTION

$F(6; -4)$  is a point on the circle with centre  $(3; -4)$ .

1. Draw a rough sketch of the circle and label  $F$ .
2. Determine the equation of the circle.
3. Does the point  $G(\frac{3}{2}; -2)$  lie on the circle?
4. Does the circle cut the  $y$ -axis? Motivate your answer.

#### SOLUTION

##### Step 1: Draw a sketch



##### Step 2: Determine the equation of the circle

Write down the general equation of a circle with centre  $(a; b)$  and substitute the coordinates  $(3; -4)$ :

$$\begin{aligned}(x - a)^2 + (y - b)^2 &= r^2 \\(x - 3)^2 + (y - (-4))^2 &= r^2 \\(x - 3)^2 + (y + 4)^2 &= r^2\end{aligned}$$

Substitute the coordinates of  $F(6; -4)$  to determine the value of  $r^2$ :

$$\begin{aligned}(6 - 3)^2 + (-4 + 4)^2 &= r^2 \\(3)^2 + (0)^2 &= r^2 \\9 &= r^2 \\\therefore (x - 3)^2 + (y + 4)^2 &= 9\end{aligned}$$

**Step 3: Determine whether or not  $G$  lies on the circle**

If  $G\left(\frac{3}{2}; -2\right)$  lies on the circle, then it will satisfy the equation of the circle:

$$\begin{aligned}\text{LHS} &= \left(\frac{3}{2} - 3\right)^2 + (-2 + 4)^2 \\ &= \left(-\frac{3}{2}\right)^2 + (2)^2 \\ &= \frac{9}{4} + 4 \\ &= \frac{9}{4} + \frac{16}{4} \\ &= \frac{25}{4}\end{aligned}$$

$$\text{RHS} = 9$$

$$\therefore \text{LHS} \neq \text{RHS}$$

Therefore  $G$  does not lie on the circle.

**Step 4: Determine the  $y$ -intercept(s)**

To determine the  $y$ -intercept(s), we let  $x = 0$ :

$$\begin{aligned}(0 - 3)^2 + (y + 4)^2 &= 9 \\ 9 + (y + 4)^2 &= 9 \\ (y + 4)^2 &= 0 \\ y + 4 &= 0 \\ \therefore y &= -4\end{aligned}$$

The circle cuts the  $y$ -axis at  $(0; -4)$ .

**Worked example 8: Equation of a circle with centre at  $(a; b)$** **QUESTION**

Determine the coordinates of the centre of the circle and the length of the radius for  $3x^2 + 6x + 3y^2 - 12y - 33 = 0$ .

**SOLUTION****Step 1: Make the coefficient of the  $x^2$  term and the  $y^2$  term equal to 1**

The coefficient of the  $x^2$  and  $y^2$  term must be 1, so we take out 3 as a common factor:

$$x^2 + 2x + y^2 - 4y - 11 = 0$$

**Step 2: Complete the square**

Take half the **coefficient of the  $x$  term**, square it; then add and subtract it from the equation.

The coefficient of the  $x$  term is 2, so then  $\left(\frac{2}{2}\right)^2 = (1)^2 = 1$ .



Take half the **coefficient of the  $y$  term**, square it; then add and subtract it from the equation.

The coefficient of the  $y$  term is  $-4$ , so then  $\left(\frac{-4}{2}\right)^2 = (-2)^2 = 4$ .

$$\begin{aligned}x^2 + 2x + y^2 - 4y - 11 &= 0 \\(x^2 + 2x + 1) - 1 + (y^2 - 4y + 4) - 4 - 11 &= 0 \\(x + 1)^2 + (y - 2)^2 - 16 &= 0 \\(x + 1)^2 + (y - 2)^2 &= 16\end{aligned}$$

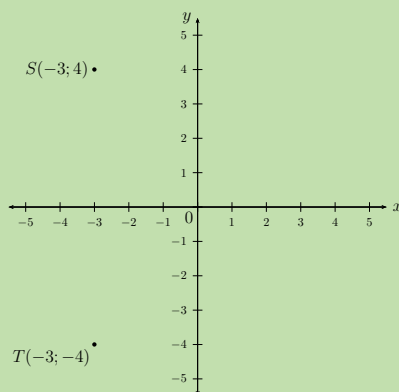
The centre of the circle is  $(-1; 2)$  and the radius is 4 units.

► See video: 293G at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)

### Worked example 9: Equation of a circle with centre at $(a; b)$

#### QUESTION

Given  $S(-3; 4)$  and  $T(-3; -4)$  on the Cartesian plane.

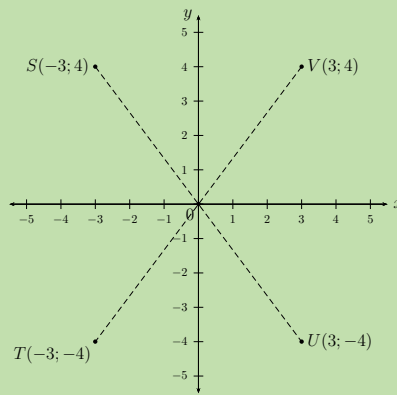


1. The points  $U$  and  $V$  are symmetrical about the origin to  $S$  and  $T$  respectively. Determine the coordinates of  $U$  and  $V$ .
2. Determine the mid-point of  $SU$ .
3. Write down the equation of the circle  $STUV$ .
4. Is  $\hat{STU} = 90^\circ$ ? Give reasons.
5. Determine the equation of the line perpendicular to  $SU$  and passing through point  $S$ .

#### SOLUTION

##### Step 1: Determine the coordinates of $U$ and $V$

For symmetry about the origin, every point  $(x; y)$  is symmetrical to  $(-x; -y)$ . So  $S(-3; 4)$  is symmetrical to  $U(3; -4)$  and  $T(-3; -4)$  is symmetrical to  $V(3; 4)$ .



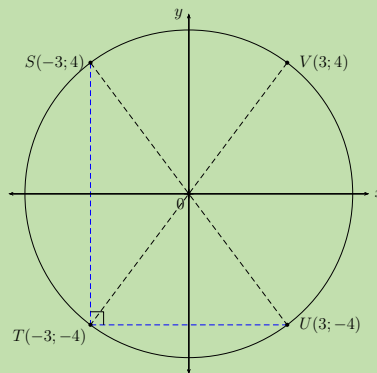
**Step 2: Determine the mid-point of  $SU$**

$$\begin{aligned}
 M(x; y) &= \left( \frac{x_U + x_S}{2}; \frac{y_U + y_S}{2} \right) \\
 &= \left( \frac{3 - 3}{2}; \frac{-4 + 4}{2} \right) \\
 &= (0; 0)
 \end{aligned}$$

The mid-point of the line  $SU$  is the origin.

**Step 3: Determine the equation of the circle**

$$\begin{aligned}
 x^2 + y^2 &= r^2 \\
 (-3)^2 + (4)^2 &= r^2 \\
 9 + 16 &= r^2 \\
 25 &= r^2 \\
 \therefore x^2 + y^2 &= 25
 \end{aligned}$$



$SU$  passes through the centre of the circle and is therefore a diameter. From Euclidean geometry, we know that the diameter of the circle subtends a right-angle at the circumference, therefore  $\hat{STU} = 90^\circ$  (angle in semi-circle).

**Step 4: Determine the equation of the line perpendicular to  $SU$  at point  $S$** 

Determine the gradient of  $SU$ :

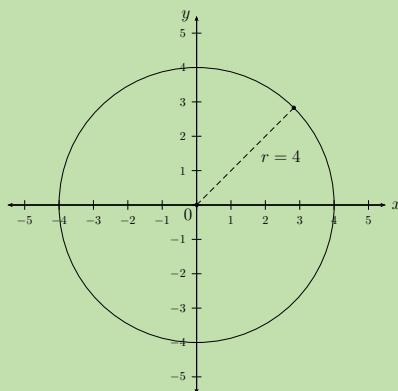
$$\begin{aligned}m_{SU} &= \frac{y_S - y_U}{x_S - x_U} \\&= \frac{4 - (-4)}{-3 - 3} \\&= \frac{8}{-6} \\&= -\frac{4}{3}\end{aligned}$$

Let the gradient of the line perpendicular to  $SU$  be  $m_P$ :

$$\begin{aligned}m_{SU} \times m_P &= -1 \\-\frac{4}{3} \times m_P &= -1 \\\therefore m_P &= \frac{3}{4} \\y - y_1 &= m(x - x_1) \\\text{Substitute } S(-3; 4): \quad y - 4 &= \frac{3}{4}(x - (-3)) \\y - 4 &= \frac{3}{4}(x + 3) \\y &= \frac{3}{4}x + \frac{25}{4}\end{aligned}$$

**Worked example 10: Equation of a circle with centre at  $(a; b)$** **QUESTION**

Given a circle with centre  $(0; 0)$  and a radius of 4 units.



1. If the circle is shifted 2 units down and 1 unit to the right, write down the equation of the shifted circle.
2. Sketch the original circle and the shifted circle on the same system of axes.

3. The shifted circle is reflected about the line  $y = x$ . Sketch the reflected circle on the same system of axes as the question above.
4. Write down the equation of the reflected circle.

### SOLUTION

#### Step 1: Write down the equation of the circle

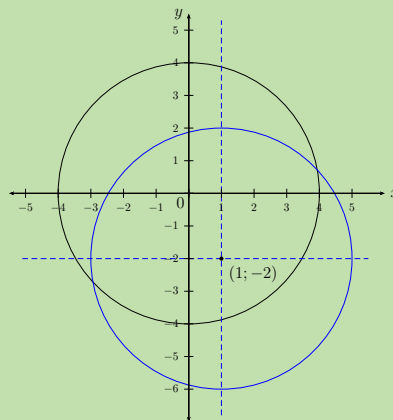
$$x^2 + y^2 = 16$$

#### Step 2: Determine the equation of the shifted circle

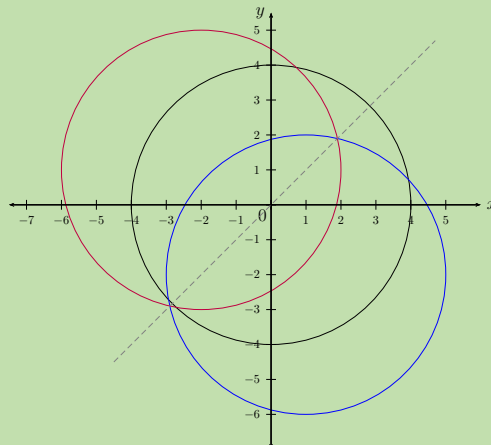
- Vertical shift: 2 units down:  $y$  is replaced with  $y + 2$
- Horizontal shift: 1 unit to the right:  $x$  is replaced with  $x - 1$

Therefore the equation of the shifted circle is  $(x - 1)^2 + (y + 2)^2 = 16$  with centre at  $(1; -2)$  and radius of 4 units.

#### Step 3: Draw a sketch of the circles



The shifted circle is reflected about the line  $y = x$ . The  $x$  and  $y$  variables are interchanged to give the circle with equation  $(y - 1)^2 + (x + 2)^2 = 16$  and centre at  $(-2; 1)$ .

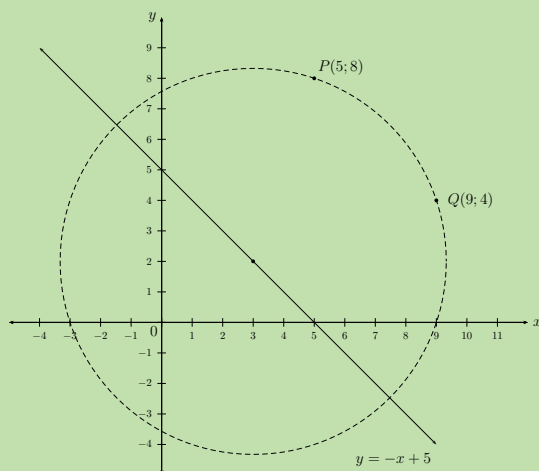


### QUESTION

A circle with centre on the line  $y = -x + 5$  passes through the points  $P(5; 8)$  and  $Q(9; 4)$ . Determine the equation of the circle.

### SOLUTION

**Step 1: Draw a rough sketch**



**Step 2: Write down the general equation of a circle**

$$(x - a)^2 + (y - b)^2 = r^2$$

Consider the line  $y = -x + 5$ . Any point on this line will have the coordinates  $(x; -x + 5)$ . Since the centre of the circle lies on the line  $y = -x + 5$ , we can write the equation of the circle as

$$\begin{aligned}(x - a)^2 + (y - (-a + 5))^2 &= r^2 \\ (x - a)^2 + (y + a - 5)^2 &= r^2\end{aligned}$$

**Step 3: Solve for the unknown variables  $a$  and  $r$**

We need two equations to solve for the two unknown variables. We substitute the two given points,  $P(5; 8)$  and  $Q(9; 4)$  and solve for  $a$  and  $r$  simultaneously:

$$\begin{aligned}\text{Substitute } P(5; 8) : \quad (5 - a)^2 + (8 + a - 5)^2 &= r^2 \\ (5 - a)^2 + (a + 3)^2 &= r^2 \\ 25 - 10a + a^2 + a^2 + 6a + 9 &= r^2 \\ 2a^2 - 4a + 34 &= r^2 \dots (1)\end{aligned}$$

$$\begin{aligned}\text{Substitute } Q(9; 4) : \quad (9 - a)^2 + (4 + a - 5)^2 &= r^2 \\ (9 - a)^2 + (a - 1)^2 &= r^2 \\ 81 - 18a + a^2 + a^2 - 2a + 1 &= r^2 \\ 2a^2 - 20a + 82 &= r^2 \dots (2)\end{aligned}$$

$$(1) - (2) : 16a - 48 = 0$$

$$16a = 48$$

$$\therefore a = 3$$

$$\text{Substitute into (2) : } r^2 = 2(3)^2 - 20(3) + 82$$

$$= 18 - 60 + 82$$

$$= 40$$

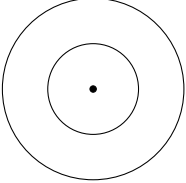
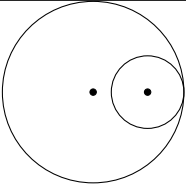
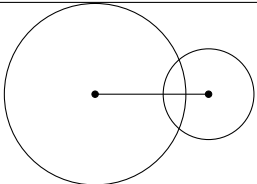
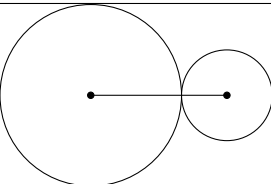
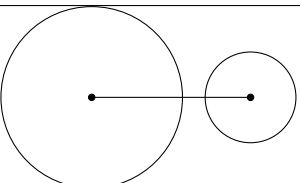
$$\text{And } b = -a + 5$$

$$= -3 + 5$$

$$= 2$$

#### Step 4: Write the final answer

The equation of the circle is  $(x - 3)^2 + (y - 2)^2 = 40$ .

|                                                                                     |                                                                                                                                          |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|    | Circles with the same centre.                                                                                                            |
|   | Circles touch internally at one point: the distance between the centres is equal to the difference between the radii.<br>$d = r_2 - r_1$ |
|  | Circles intersect at two points: the distance between the centres is less than the sum of the radii.<br>$d < r_1 + r_2$                  |
|  | Circles touch externally at one point: the distance between the centres is equal to the sum of the radii.<br>$d = r_1 + r_2$             |
|  | Circles do not touch or intersect: the distance between the centres is greater than the sum of the radii.<br>$d > r_1 + r_2$             |

**Exercise 7 – 4: Equation of a circle with centre at  $(a; b)$** 

1. Determine whether or not each of the following equations represents a circle. If not, give a reason.

a)  $x^2 + y^2 + 6y - 10 = 0$

d)  $x^2 - 4x = \sqrt{21} + 5y + y^2$

b)  $3x^2 - 35 + 3y^2 = 9y$

e)  $3\sqrt{7} - x^2 - y^2 + 6y - 8x = 0$

c)  $40 = x^2 + 2x + 4y^2$

f)  $(x - 1)^2 + (y + 2)^2 + 9 = 0$

2. Write down the equation of the circle:

a) with centre  $(0; 4)$  and a radius of 3 units.

b) such that  $r = 5$  and the centre is the origin.

c) with centre  $(-2; 3)$  and passing through the point  $(4; 5)$ .

d) with centre  $(p; -q)$  and  $r = \sqrt{6}$ .

e) with  $r = \sqrt{10}$  and centre  $(-\frac{1}{2}; \frac{3}{2})$  and

f) with centre  $(1; -5)$  and passing through the origin.

3. Determine the centre and the length of the radius for the following circles:

a)  $x^2 = 21 - y^2 + 4y$

e)  $5 - x^2 - 6x - 8y - y^2 = 0$

b)  $y^2 + x + x^2 - \frac{15}{4} = 0$

f)  $x^2 - \frac{2}{3}x + y^2 - 4y = \frac{35}{9}$

c)  $x^2 - 4x + y^2 + 2y - 5 = 0$

g)  $16x + 2y^2 - 20y + 2x^2 + 42 = 0$

d)  $x^2 + y^2 - 6y + 2x - 15 = 0$

h)  $6x - 6y - x^2 - y^2 = 6$

4. A circle cuts the  $x$ -axis at  $R(-2; 0)$  and  $S(2; 0)$ . If  $r = \sqrt{20}$  units, determine the possible equation(s) of the circle. Draw a sketch.

5.  $P(1; 2)$  and  $Q(-5; -6)$  are points on a circle such that  $PQ$  is a diameter. Determine the equation of the circle.

6. A circle with centre  $N(4; 4)$  passes through the points  $K(1; 6)$  and  $L(6; 7)$ .

a) Determine the equation of the circle.

b) Determine the coordinates of  $M$ , the mid-point of  $KL$ .

c) Show that  $MN \perp KL$ .

d) If  $P$  is a point on the circle such that  $LP$  is a diameter, determine the coordinates of  $P$ .

e) Determine the equation of the line  $LP$ .

7. A circle passes through the point  $A(7; -4)$  and  $B(-5; -2)$ . If its centre lies on the line  $y + 5 = 2x$ , determine the equation of the circle.

8. A circle with centre  $(0; 0)$  passes through the point  $T(3; 5)$ .

a) Determine the equation of the circle.

b) If the circle is shifted 2 units to the right and 3 units down, determine the new equation of the circle.

c) Draw a sketch of the original circle and the shifted circle on the same system of axes.

d) On the same system of axes as the previous question, draw a sketch of the shifted circle reflected about the  $x$ -axis. State the coordinates of the centre of this circle.

9. Determine whether the circle  $x^2 - 4x + y^2 - 6y + 9 = 0$  cuts, touches or does not intersect the  $x$ -axis and the  $y$ -axis.
10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |                          |                          |                          |                          |                          |                          |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. <a href="#">295Q</a> | 1b. <a href="#">295R</a> | 1c. <a href="#">295S</a> | 1d. <a href="#">295T</a> | 1e. <a href="#">295V</a> | 1f. <a href="#">295W</a> |
| 2a. <a href="#">295X</a> | 2b. <a href="#">295Y</a> | 2c. <a href="#">295Z</a> | 2d. <a href="#">2962</a> | 2e. <a href="#">2963</a> | 2f. <a href="#">2964</a> |
| 3a. <a href="#">2965</a> | 3b. <a href="#">2966</a> | 3c. <a href="#">2967</a> | 3d. <a href="#">2968</a> | 3e. <a href="#">2969</a> | 3f. <a href="#">296B</a> |
| 3g. <a href="#">296C</a> | 3h. <a href="#">296D</a> | 4. <a href="#">296F</a>  | 5. <a href="#">296G</a>  | 6. <a href="#">296H</a>  | 7. <a href="#">296J</a>  |
| 8. <a href="#">296K</a>  | 9. <a href="#">296M</a>  |                          |                          |                          |                          |



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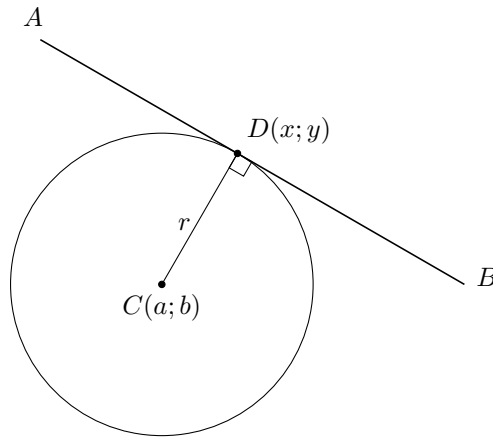
## 7.3 Equation of a tangent to a circle

EMCHW

### Investigation:

- On a suitable system of axes, draw the circle  $x^2 + y^2 = 20$  with centre at  $O(0; 0)$ .
- Plot the point  $T(2; 4)$ .
- Plot the point  $P(0; 5)$ . Draw  $PT$  and extend the line so that it cuts the positive  $x$ -axis.
- Measure  $\hat{O}TP$ .
- Determine the gradient of the radius  $OT$ .
- Determine the gradient of  $PT$ .
- Prove that  $PT \perp OT$ .
- Plot the point  $S(2; -4)$  and join  $OS$ .
- Draw a tangent to the circle at  $S$ .
- Measure the angle between  $OS$  and the tangent line at  $S$ .
- Make a conjecture about the angle between the radius and the tangent to a circle at a point on the circle.
- Complete the sentence: the product of the ..... of the radius and the gradient of the ..... is equal to .....





A circle with centre  $C(a; b)$  and a radius of  $r$  units is shown in the diagram above.  $D(x; y)$  is a point on the circumference and the equation of the circle is:

$$(x - a)^2 + (y - b)^2 = r^2$$

A tangent is a straight line that touches the circumference of a circle at only one place.

The tangent line  $AB$  touches the circle at  $D$ .

The radius of the circle  $CD$  is perpendicular to the tangent  $AB$  at the point of contact  $D$ .

$$CD \perp AB$$

$$\text{and } \hat{CDA} = \hat{CDB} = 90^\circ$$

The product of the gradient of the radius and the gradient of the tangent line is equal to  $-1$ .

$$m_{CD} \times m_{AB} = -1$$

### How to determine the equation of a tangent:

1. Determine the equation of the circle and write it in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

2. From the equation, determine the coordinates of the centre of the circle  $(a; b)$ .
3. Determine the gradient of the radius:

$$m_{CD} = \frac{y_2 - y_1}{x_2 - x_1}$$

4. The radius is perpendicular to the tangent of the circle at a point  $D$  so:

$$m_{AB} = -\frac{1}{m_{CD}}$$

5. Write down the gradient-point form of a straight line equation and substitute  $m_{AB}$  and the coordinates of  $D$ . Make  $y$  the subject of the equation.

$$y - y_1 = m(x - x_1)$$

### Worked example 12: Equation of a tangent to a circle

#### QUESTION

Determine the equation of the tangent to the circle  $x^2 + y^2 - 2y + 6x - 7 = 0$  at the point  $F(-2; 5)$ .

#### SOLUTION

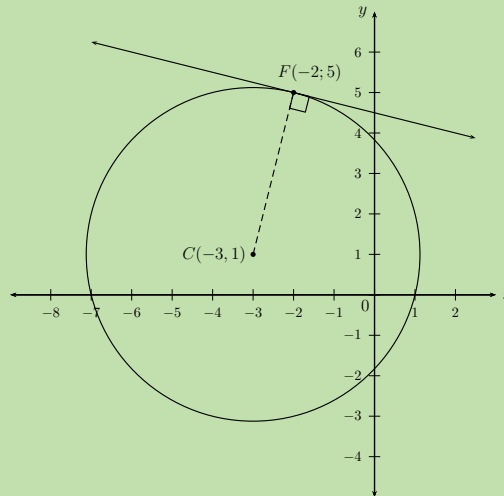
**Step 1: Write the equation of the circle in the form  $(x - a)^2 + (y - b)^2 = r^2$**

Use the method of completing the square:

$$\begin{aligned}x^2 + y^2 - 2y + 6x - 7 &= 0 \\x^2 + 6x + y^2 - 2y &= 7 \\(x^2 + 6x + 9) - 9 + (y^2 - 2y + 1) - 1 &= 7 \\(x + 3)^2 + (y - 1)^2 &= 17\end{aligned}$$

**Step 2: Draw a sketch**

The centre of the circle is  $(-3; 1)$  and the radius is  $\sqrt{17}$  units.



**Step 3: Determine the gradient of the radius  $CF$**

$$\begin{aligned}m_{CF} &= \frac{y_2 - y_1}{x_2 - x_1} \\&= \frac{5 - 1}{-2 - (-3)} \\&= 4\end{aligned}$$

**Step 4: Determine the gradient of the tangent**

Let the gradient of the tangent line be  $m$ .

$$\begin{aligned}m_{CF} \times m &= -1 \\4 \times m &= -1 \\\therefore m &= -\frac{1}{4}\end{aligned}$$

**Step 5: Determine the equation of the tangent to the circle**

Write down the gradient-point form of a straight line equation and substitute  $m = -\frac{1}{4}$  and  $F(-2; 5)$ .

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = -\frac{1}{4}(x - x_1)$$

$$\text{Substitute } F(-2; 5) : \quad y - 5 = -\frac{1}{4}(x - (-2))$$

$$y - 5 = -\frac{1}{4}(x + 2)$$

$$y = -\frac{1}{4}x - \frac{1}{2} + 5$$

$$= -\frac{1}{4}x + \frac{9}{2}$$

**Step 6: Write the final answer**

The equation of the tangent to the circle at  $F$  is  $y = -\frac{1}{4}x + \frac{9}{2}$ .

**Worked example 13: Equation of a tangent to a circle****QUESTION**

The straight line  $y = x + 4$  cuts the circle  $x^2 + y^2 = 26$  at  $P$  and  $Q$ .

1. Calculate the coordinates of  $P$  and  $Q$ .
2. Sketch the circle and the straight line on the same system of axes. Label points  $P$  and  $Q$ .
3. Determine the coordinates of  $H$ , the mid-point of chord  $PQ$ .
4. If  $O$  is the centre of the circle, show that  $PQ \perp OH$ .
5. Determine the equations of the tangents to the circle at  $P$  and  $Q$ .
6. Determine the coordinates of  $S$ , the point where the two tangents intersect.
7. Show that  $S$ ,  $H$  and  $O$  are on a straight line.

**SOLUTION****Step 1: Determine the coordinates of  $P$  and  $Q$** 

Substitute the straight line  $y = x + 4$  into the equation of the circle and solve for  $x$ :

$$x^2 + y^2 = 26$$

$$x^2 + (x + 4)^2 = 26$$

$$x^2 + x^2 + 8x + 16 = 26$$

$$2x^2 + 8x - 10 = 0$$

$$x^2 + 4x - 5 = 0$$

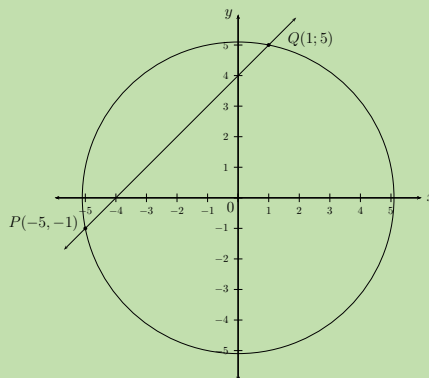
$$(x - 1)(x + 5) = 0$$

$$\therefore x = 1 \text{ or } x = -5$$

$$\text{If } x = 1 \quad y = 1 + 4 = 5$$

$$\text{If } x = -5 \quad y = -5 + 4 = -1$$

This gives the points  $P(-5; -1)$  and  $Q(1; 5)$ .

**Step 2: Draw a sketch****Step 3: Determine the coordinates of the mid-point  $H$** 

$$\begin{aligned}
 H(x; y) &= \left( \frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right) \\
 &= \left( \frac{1 - 5}{2}; \frac{5 - 1}{2} \right) \\
 &= \left( \frac{-4}{2}; \frac{4}{2} \right) \\
 &= (-2; 2)
 \end{aligned}$$

**Step 4: Show that  $OH$  is perpendicular to  $PQ$** 

We need to show that the product of the two gradients is equal to  $-1$ . From the given equation of  $PQ$ , we know that  $m_{PQ} = 1$ .

$$\begin{aligned}
 m_{OH} &= \frac{2 - 0}{-2 - 0} \\
 &= -1 \\
 m_{PQ} \times m_{OH} &= -1 \\
 \therefore PQ \perp OH
 \end{aligned}$$

**Step 5: Determine the equations of the tangents at  $P$  and  $Q$** **Tangent at  $P$ :**

Determine the gradient of the radius  $OP$ :

$$\begin{aligned}
 m_{OP} &= \frac{-1 - 0}{-5 - 0} \\
 &= \frac{1}{5}
 \end{aligned}$$

The tangent of a circle is perpendicular to the radius, therefore we can write:

$$\begin{aligned}
 \frac{1}{5} \times m_P &= -1 \\
 \therefore m_P &= -5
 \end{aligned}$$

Substitute  $m_P = -5$  and  $P(-5; -1)$  into the equation of a straight line.

$$y - y_1 = -5(x - x_1)$$

Substitute  $P(-5; -1)$  :

$$\begin{aligned}
 y + 1 &= -5(x + 5) \\
 y &= -5x - 25 - 1 \\
 &= -5x - 26
 \end{aligned}$$

**Tangent at  $Q$ :**

Determine the gradient of the radius  $OQ$ :

$$\begin{aligned}
 m_{OQ} &= \frac{5 - 0}{1 - 0} \\
 &= 5
 \end{aligned}$$

The tangent of a circle is perpendicular to the radius, therefore we can write:

$$\begin{aligned}
 5 \times m_Q &= -1 \\
 \therefore m_Q &= -\frac{1}{5}
 \end{aligned}$$

Substitute  $m_Q = -\frac{1}{5}$  and  $Q(1; 5)$  into the equation of a straight line.

$$y - y_1 = -\frac{1}{5}(x - x_1)$$

Substitute  $Q(1; 5)$  :

$$\begin{aligned}
 y - 5 &= -\frac{1}{5}(x - 1) \\
 y &= -\frac{1}{5}x + \frac{1}{5} + 5 \\
 &= -\frac{1}{5}x + \frac{26}{5}
 \end{aligned}$$

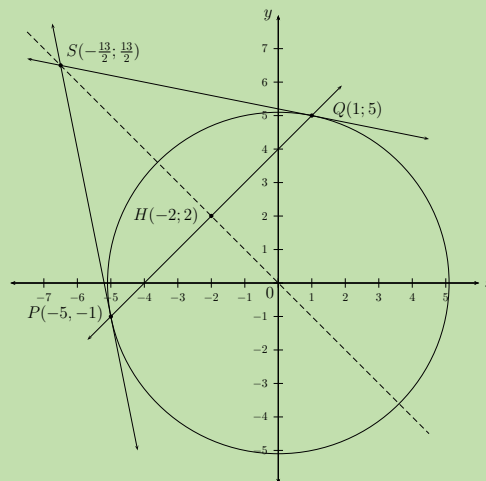
The equations of the tangents are  $y = -5x - 26$  and  $y = -\frac{1}{5}x + \frac{26}{5}$ .

**Step 6: Determine the coordinates of  $S$**

Equate the two linear equations and solve for  $x$ :

$$\begin{aligned} -5x - 26 &= -\frac{1}{5}x + \frac{26}{5} \\ -25x - 130 &= -x + 26 \\ -24x &= 156 \\ x &= -\frac{156}{24} \\ &= -\frac{13}{2} \\ \text{If } x &= -\frac{13}{2} \quad y = -5\left(-\frac{13}{2}\right) - 26 \\ &= \frac{65}{2} - 26 \\ &= \frac{13}{2} \end{aligned}$$

This gives the point  $S\left(-\frac{13}{2}; \frac{13}{2}\right)$ .



**Step 7: Show that  $S$ ,  $H$  and  $O$  are on a straight line**

We need to show that there is a constant gradient between any two of the three points. We have already shown that  $PQ$  is perpendicular to  $OH$ , so we expect the gradient of the line through  $S$ ,  $H$  and  $O$  to be  $-1$ .

$$\begin{aligned} m_{SH} &= \frac{\frac{13}{2} - 2}{-\frac{13}{2} + 2} \\ &= -1 \\ m_{SO} &= \frac{\frac{13}{2} - 0}{-\frac{13}{2} - 0} \\ &= -1 \end{aligned}$$

Therefore  $S$ ,  $H$  and  $O$  all lie on the line  $y = -x$ .

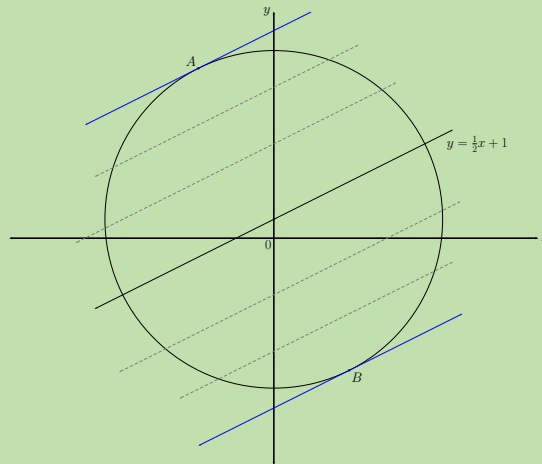
### Worked example 14: Equation of a tangent to a circle

#### QUESTION

Determine the equations of the tangents to the circle  $x^2 + (y - 1)^2 = 80$ , given that both are parallel to the line  $y = \frac{1}{2}x + 1$ .

#### SOLUTION

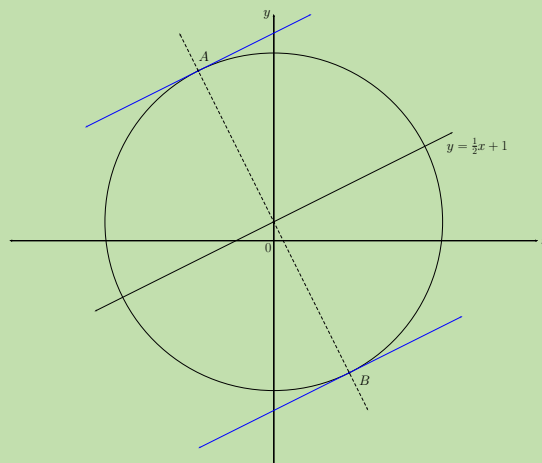
**Step 1: Draw a sketch**



The tangents to the circle, parallel to the line  $y = \frac{1}{2}x + 1$ , must have a gradient of  $\frac{1}{2}$ . From the sketch we see that there are two possible tangents.

**Step 2: Determine the coordinates of  $A$  and  $B$**

To determine the coordinates of  $A$  and  $B$ , we must find the equation of the line perpendicular to  $y = \frac{1}{2}x + 1$  and passing through the centre of the circle. This perpendicular line will cut the circle at  $A$  and  $B$ .



$$y = \frac{1}{2}x + 1$$

$$\therefore m = \frac{1}{2}$$

$$m_{\perp} = -\frac{1}{m} = -2$$

$$\therefore y = -2x + 1$$

Notice that the line passes through the centre of the circle.

To determine the coordinates of  $A$  and  $B$ , we substitute the straight line  $y = -2x + 1$  into the equation of the circle and solve for  $x$ :

$$x^2 + (y - 1)^2 = 80$$

$$x^2 + (-2x + 1 - 1)^2 = 80$$

$$x^2 + 4x^2 = 80$$

$$5x^2 = 80$$

$$x^2 = 16$$

$$\therefore x = \pm 4$$

$$\text{If } x = 4 \quad y = -2(4) + 1 = -7$$

$$\text{If } x = -4 \quad y = -2(-4) + 1 = 9$$

This gives the points  $A(-4; 9)$  and  $B(4; -7)$ .

### Step 3: Determine the equations of the tangents to the circle

**Tangent at  $A$ :**

$$y - y_1 = \frac{1}{2}(x - x_1)$$

$$y - 9 = \frac{1}{2}(x + 4)$$

$$y = \frac{1}{2}x + 11$$

**Tangent at  $B$ :**

$$y - y_1 = \frac{1}{2}(x - x_1)$$

$$y + 7 = \frac{1}{2}(x - 4)$$

$$y = \frac{1}{2}x - 9$$

The equation of the tangent at point  $A$  is  $y = \frac{1}{2}x + 11$  and the equation of the tangent at point  $B$  is  $y = \frac{1}{2}x - 9$ .

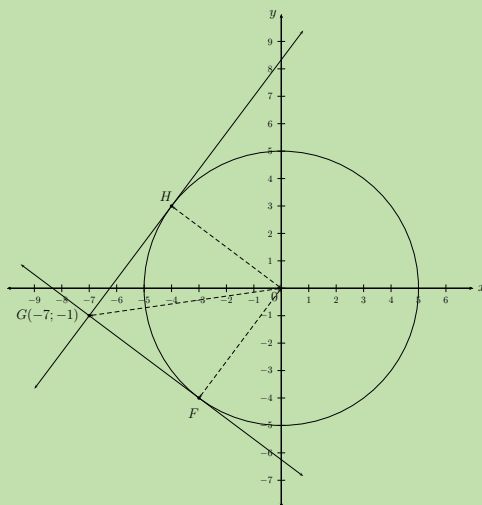
### Worked example 15: Equation of a tangent to a circle

#### QUESTION

Determine the equations of the tangents to the circle  $x^2 + y^2 = 25$ , from the point  $G(-7; -1)$  outside the circle.

#### SOLUTION

**Step 1: Draw a sketch**



**Step 2: Consider where the two tangents will touch the circle**

Let the two tangents from  $G$  touch the circle at  $F$  and  $H$ .

$$OF = OH = 5 \text{ units} \quad (\text{equal radii})$$

$$\begin{aligned} OG &= \sqrt{(0+7)^2 + (0+1)^2} \\ &= \sqrt{50} \end{aligned}$$

$$GF = \sqrt{(x+7)^2 + (y+1)^2}$$

$$\therefore GF^2 = (x+7)^2 + (y+1)^2$$

$$\text{And } \hat{GFO} = \hat{HGO} = 90^\circ$$

Consider  $\triangle GFO$  and apply the theorem of Pythagoras:

$$GF^2 + OF^2 = OG^2$$

$$(x+7)^2 + (y+1)^2 + 5^2 = (\sqrt{50})^2$$

$$x^2 + 14x + 49 + y^2 + 2y + 1 + 25 = 50$$

$$x^2 + 14x + y^2 + 2y + 25 = 0 \dots\dots (1)$$

Substitute  $y^2 = 25 - x^2$  into equation (1)

$$x^2 + 14x + (25 - x^2) + 2(\sqrt{25 - x^2}) + 25 = 0$$

$$14x + 50 = -2(\sqrt{25 - x^2})$$

$$7x + 25 = -\sqrt{25 - x^2}$$



$$\text{Square both sides: } (7x + 25)^2 = \left(-\sqrt{25 - x^2}\right)^2$$

$$49x^2 + 350x + 625 = 25 - x^2$$

$$50x^2 + 350x + 600 = 0$$

$$x^2 + 7x + 12 = 0$$

$$(x + 3)(x + 4) = 0$$

$$\therefore x = -3 \text{ or } x = -4$$

$$\text{At } F : x = -3 \quad y = -\sqrt{25 - (-3)^2} = -\sqrt{16} = -4$$

$$\text{At } H : x = -4 \quad y = \sqrt{25 - (-4)^2} = \sqrt{9} = 3$$

**Note:** from the sketch we see that  $F$  must have a negative  $y$ -coordinate, therefore we take the negative of the square root. Similarly,  $H$  must have a positive  $y$ -coordinate, therefore we take the positive of the square root.

This gives the points  $F(-3; -4)$  and  $H(-4; 3)$ .

**Tangent at  $F$ :**

$$\begin{aligned} m_{FG} &= \frac{-1 + 4}{-7 + 3} \\ &= -\frac{3}{4} \end{aligned}$$

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = -\frac{3}{4}(x - x_1)$$

$$y + 1 = -\frac{3}{4}(x + 7)$$

$$y = -\frac{3}{4}x - \frac{21}{4} - 1$$

$$y = -\frac{3}{4}x - \frac{25}{4}$$

**Tangent at  $H$ :**

$$\begin{aligned} m_{HG} &= \frac{-1 - 3}{-7 + 4} \\ &= \frac{4}{3} \end{aligned}$$

$$y + 1 = \frac{4}{3}(x + 7)$$

$$y = \frac{4}{3}x + \frac{28}{3} - 1$$

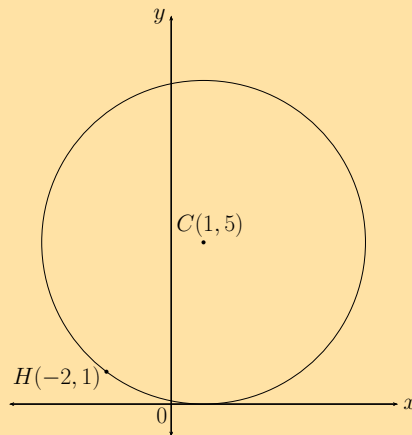
$$y = \frac{4}{3}x + \frac{25}{3}$$

**Step 3: Write the final answer**

The equations of the tangents to the circle are  $y = -\frac{3}{4}x - \frac{25}{4}$  and  $y = \frac{4}{3}x + \frac{25}{3}$ .

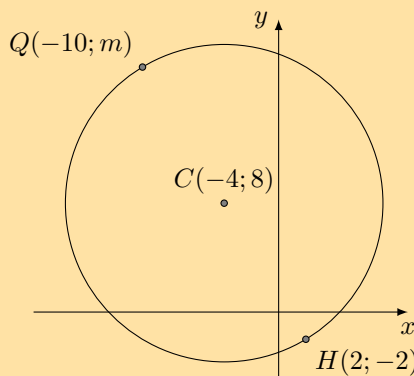
### Exercise 7 – 5: Equation of a tangent to a circle

1. a) A circle with centre  $(8; -7)$  and the point  $(5; -5)$  on the circle are given. Determine the gradient of the radius.  
b) Determine the gradient of the tangent to the circle at the point  $(5; -5)$ .
2. Given the equation of the circle:  $(x + 4)^2 + (y + 8)^2 = 136$   
a) Find the gradient of the radius at the point  $(2; 2)$  on the circle.  
b) Determine the gradient of the tangent to the circle at the point  $(2; 2)$ .
3. Given a circle with the central coordinates  $(a; b) = (-9; 6)$ . Determine the equation of the tangent to the circle at the point  $(-2; 5)$ .
4. Given the diagram below:



Determine the equation of the tangent to the circle with centre  $C$  at point  $H$ .

5. Given the point  $P(2; -4)$  on the circle  $(x - 4)^2 + (y + 5)^2 = 5$ . Find the equation of the tangent at  $P$ .
6.  $C(-4; 8)$  is the centre of the circle passing through  $H(2; -2)$  and  $Q(-10; m)$ .



- a) Determine the equation of the circle.
- b) Determine the value of  $m$ .
- c) Determine the equation of the tangent to the circle at point  $Q$ .

7. The straight line  $y = x + 2$  cuts the circle  $x^2 + y^2 = 20$  at  $P$  and  $Q$ .
- Calculate the coordinates of  $P$  and  $Q$ .
  - Determine the length of  $PQ$ .
  - Determine the coordinates of  $M$ , the mid-point of chord  $PQ$ .
  - If  $O$  is the centre of the circle, show that  $PQ \perp OM$ .
  - Determine the equations of the tangents to the circle at  $P$  and  $Q$ .
  - Determine the coordinates of  $S$ , the point where the two tangents intersect.
  - Show that  $PS = QS$ .
  - Determine the equations of the two tangents to the circle, both parallel to the line  $y + 2x = 4$ .
8. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 296N   2. 296P   3. 296Q   4. 296R   5. 296S   6. 296T  
7. 296V



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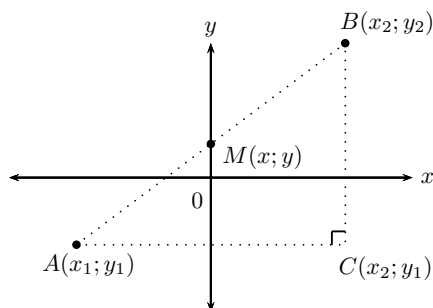


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## 7.4 Summary

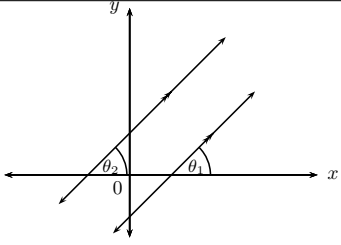
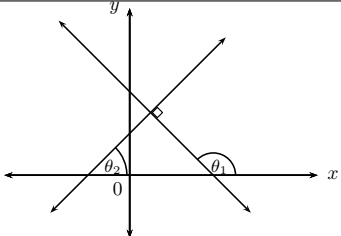
EMCHX

► See video: 294R at [www.everythingmaths.co.za](http://www.everythingmaths.co.za)



|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| Theorem of Pythagoras:       | $AB^2 = AC^2 + BC^2$                                                             |
| Distance formula:            | $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$                                      |
| Gradient:                    | $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$ or $m_{AB} = \frac{y_1 - y_2}{x_1 - x_2}$ |
| Mid-point of a line segment: | $M(x; y) = \left( \frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$              |
| Points on a straight line:   | $m_{AB} = m_{AM} = m_{MB}$                                                       |

| Straight line equations  | Formulae                                                |
|--------------------------|---------------------------------------------------------|
| Two-point form:          | $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ |
| Gradient-point form:     | $y - y_1 = m(x - x_1)$                                  |
| Gradient-intercept form: | $y = mx + c$                                            |
| Horizontal lines:        | $y = k$                                                 |
| Vertical lines           | $x = k$                                                 |

|                     |                                                                                   |                       |                                  |
|---------------------|-----------------------------------------------------------------------------------|-----------------------|----------------------------------|
| Parallel lines      |  | $m_1 = m_2$           | $\theta_1 = \theta_2$            |
| Perpendicular lines |  | $m_1 \times m_2 = -1$ | $\theta_1 = 90^\circ + \theta_2$ |

- Inclination of a straight line: the gradient of a straight line is equal to the tangent of the angle formed between the line and the positive direction of the  $x$ -axis.

$$m = \tan \theta \quad \text{for } 0^\circ \leq \theta < 180^\circ$$

- Equation of a circle with centre at the origin:  
If  $P(x; y)$  is a point on a circle with centre  $O(0; 0)$  and radius  $r$ , then the equation of the circle is:

$$x^2 + y^2 = r^2$$

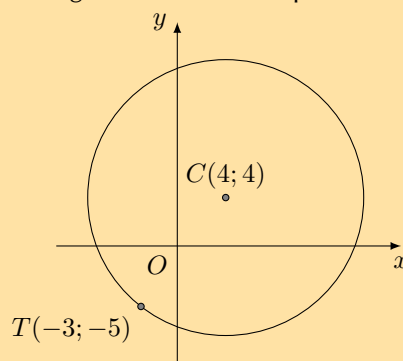
- General equation of a circle with centre at  $(a; b)$ :  
If  $P(x; y)$  is a point on a circle with centre  $C(a; b)$  and radius  $r$ , then the equation of the circle is:

$$(x - a)^2 + (y - b)^2 = r^2$$

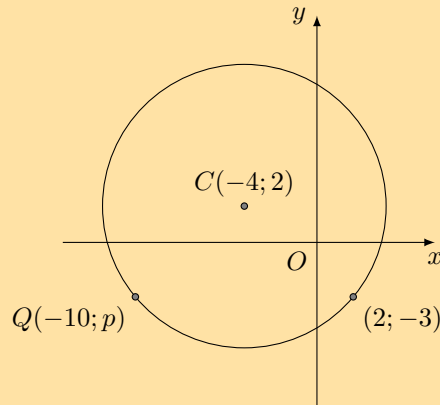
- A tangent is a straight line that touches the circumference of a circle at only one point.
- The radius of a circle is perpendicular to the tangent at the point of contact.

## Exercise 7 – 6: End of chapter exercises

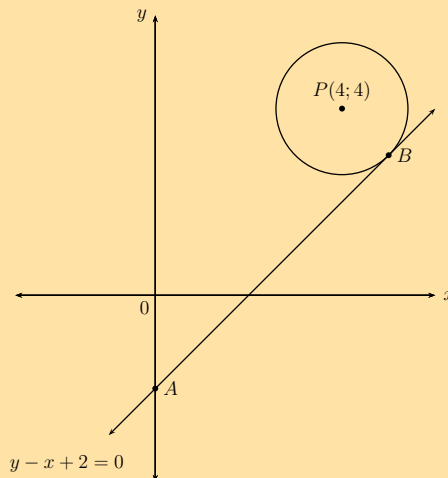
1. Find the equation of the circle:
  - a) with centre  $(0; 5)$  and radius 5
  - b) with centre  $(2; 0)$  and radius 4
  - c) with centre  $(-5; 7)$  and radius 18
  - d) with centre  $(-2; 0)$  and diameter 6
  - e) with centre  $(-5; -3)$  and radius  $\sqrt{3}$
2.
  - a) Find the equation of the circle with centre  $(2; 1)$  which passes through  $(4; 1)$ .
  - b) Where does it cut the line  $y = x + 1$ ?
3.
  - a) Find the equation of the circle with centre  $(-3; -2)$  which passes through  $(1; -4)$ .
  - b) Find the equation of the circle with centre  $(3; 1)$  which passes through  $(2; 5)$ .
4. Find the centre and radius of the following circles:
  - a)  $(x + 9)^2 + (y - 6)^2 = 36$
  - b)  $\frac{1}{2}(x - 2)^2 + \frac{1}{2}(y - 9)^2 = 1$
  - c)  $(x + 5)^2 + (y + 7)^2 = 12$
  - d)  $x^2 + (y + 4)^2 = 23$
  - e)  $3(x - 2)^2 + 3(y + 3)^2 = 12$
5. Find the  $x$  and  $y$  intercepts of the following graphs:
  - a)  $x^2 + (y - 6)^2 = 100$
  - b)  $(x + 4)^2 + y^2 = 16$
6. Find the centre and radius of the following circles:
  - a)  $x^2 + 6x + y^2 - 12y = -20$
  - b)  $x^2 + 4x + y^2 - 8y = 0$
  - c)  $x^2 + y^2 + 8y = 7$
  - d)  $x^2 - 6x + y^2 = 16$
  - e)  $x^2 - 5x + y^2 + 3y = -\frac{3}{4}$
  - f)  $x^2 - 6nx + y^2 + 10ny = 9n^2$
7.
  - a) Find the gradient of the radius between the point  $(4; 5)$  on the circle and its centre  $(-8; 4)$ .
  - b) Find the gradient line tangent to the circle at the point  $(4; 5)$ .
8.
  - a) Given  $(x - 1)^2 + (y - 7)^2 = 10$ , determine the value(s) of  $x$  if  $(x; 4)$  lies on the circle.
  - b) Find the gradient of the tangent to the circle at the point  $(2; 4)$ .
9. Given a circle with the central coordinates  $(a; b) = (-2; -2)$ . Determine the equation of the tangent line of the circle at the point  $(-1; 3)$ .
10. Find the equation of the tangent to the circle at point  $T$ .



11.  $M(-2; -5)$  is a point on the circle  $x^2 + y^2 + 18y + 61 = 0$ . Determine the equation of the tangent at  $M$ .
12.  $C(-4; 2)$  is the centre of the circle passing through  $(2; -3)$  and  $Q(-10; p)$ .



- Find the equation of the circle given.
  - Determine the value of  $p$ .
  - Determine the equation of the tangent to the circle at point  $Q$ .
13. Find the equation of the tangent to each circle:
- $x^2 + y^2 = 17$  at the point  $(1; 4)$
  - $x^2 + y^2 = 25$  at the point  $(3; 4)$
  - $(x + 1)^2 + (y - 2)^2 = 25$  at the point  $(3; 5)$
  - $(x - 2)^2 + (y - 1)^2 = 13$  at the point  $(5; 3)$
14. Determine the equations of the tangents to the circle  $x^2 + y^2 = 50$ , given that both lines have an angle of inclination of  $45^\circ$ .
15. The circle with centre  $P(4; 4)$  has a tangent  $AB$  at point  $B$ . The equation of  $AB$  is  $y - x + 2 = 0$  and  $A$  lies on the  $y$ -axis.



- Determine the equation of  $PB$ .
- Determine the coordinates of  $B$ .
- Determine the equation of the circle.
- Describe in words how the circle must be shifted so that  $P$  is at the origin.

- e) If the length of  $PB$  is tripled and the circle is shifted 2 units to the right and 1 unit up, determine the equation of the new circle.
- f) The equation of a circle with centre  $A$  is  $x^2 + y^2 + 5 = 16x + 8y - 30$  and the equation of a circle with centre  $B$  is  $5x^2 + 5y^2 = 25$ . Prove that the two circles touch each other.

16. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- |                          |                          |                           |                           |                           |                           |
|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| 1a. <a href="#">296W</a> | 1b. <a href="#">296X</a> | 1c. <a href="#">296Y</a>  | 1d. <a href="#">296Z</a>  | 1e. <a href="#">2972</a>  | 2. <a href="#">2973</a>   |
| 3. <a href="#">2974</a>  | 4a. <a href="#">2975</a> | 4b. <a href="#">2976</a>  | 4c. <a href="#">2977</a>  | 4d. <a href="#">2978</a>  | 4e. <a href="#">2979</a>  |
| 5a. <a href="#">297B</a> | 5b. <a href="#">297C</a> | 6a. <a href="#">297D</a>  | 6b. <a href="#">297F</a>  | 6c. <a href="#">297G</a>  | 6d. <a href="#">297H</a>  |
| 6e. <a href="#">297J</a> | 6f. <a href="#">297K</a> | 7. <a href="#">297M</a>   | 8. <a href="#">297N</a>   | 9. <a href="#">297P</a>   | 10. <a href="#">297Q</a>  |
| 11. <a href="#">297R</a> | 12. <a href="#">297S</a> | 13a. <a href="#">297T</a> | 13b. <a href="#">297V</a> | 13c. <a href="#">297W</a> | 13d. <a href="#">297X</a> |
| 14. <a href="#">297Y</a> | 15. <a href="#">297Z</a> |                           |                           |                           |                           |



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## *Euclidean geometry*

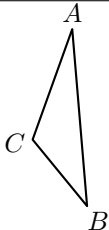
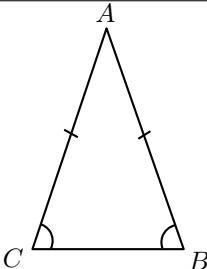
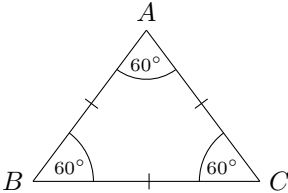
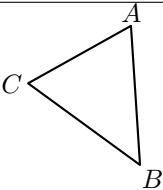
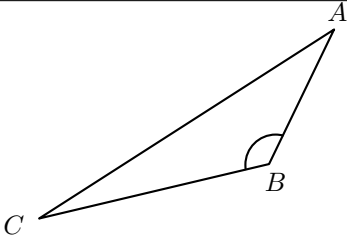
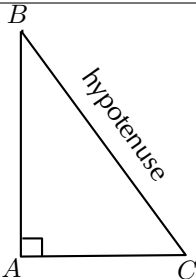
|     |                             |     |
|-----|-----------------------------|-----|
| 8.1 | <i>Revision</i>             | 312 |
| 8.2 | <i>Ratio and proportion</i> | 319 |
| 8.3 | <i>Polygons</i>             | 323 |
| 8.4 | <i>Triangles</i>            | 327 |
| 8.5 | <i>Similarity</i>           | 334 |
| 8.6 | <i>Pythagorean theorem</i>  | 348 |
| 8.7 | <i>Summary</i>              | 353 |

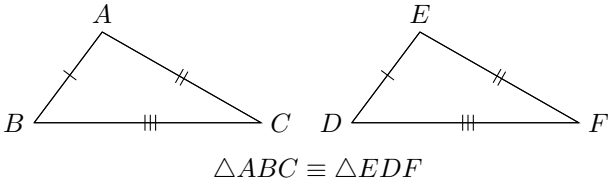
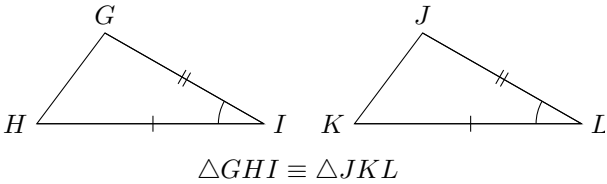
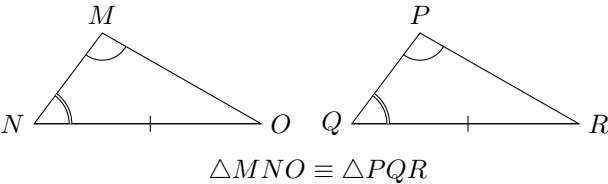
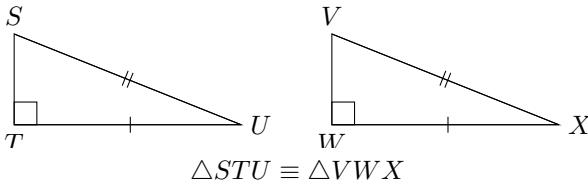
8.1 Revision

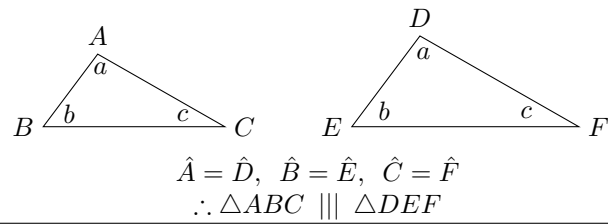
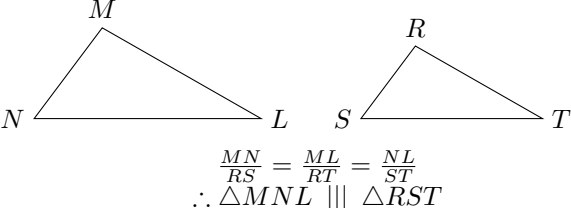
EMCHY

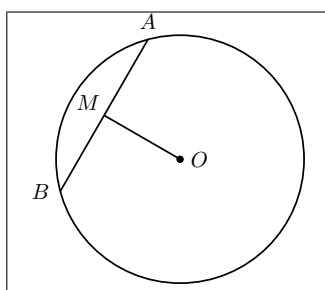
Types of triangles

EMCHZ

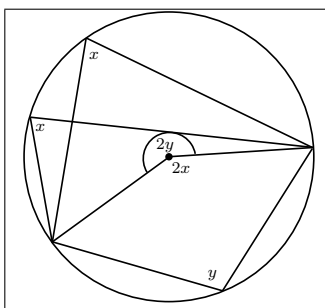
| Name          | Diagram                                                                             | Properties                                                                         |
|---------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Scalene       |    | All sides and angles are different.                                                |
| Isosceles     |    | Two sides are equal in length. The angles opposite the equal sides are also equal. |
| Equilateral   |   | All three sides are equal in length and all three angles are equal.                |
| Acute-angled  |  | Each of the three interior angles is less than 90°.                                |
| Obtuse-angled |  | One interior angle is greater than 90°.                                            |
| Right-angled  |  | One interior angle is 90°.                                                         |

| Condition                        | Diagram                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| SSS<br>(side, side, side)        |  $\triangle ABC \equiv \triangle EDF$  |
| SAS<br>(side, incl. angle, side) |  $\triangle GHI \equiv \triangle JKL$  |
| AAS<br>(angle, angle, side)      |  $\triangle MNO \equiv \triangle PQR$  |
| RHS<br>(90°, hypotenuse, side)   |  $\triangle STU \equiv \triangle VWX$ |

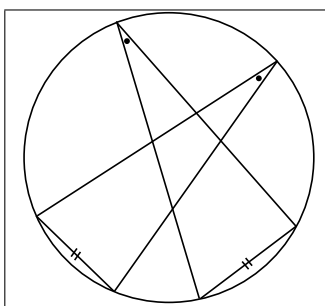
| Condition                    | Diagram                                                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AAA<br>(angle, angle, angle) |  $\hat{A} = \hat{D}, \hat{B} = \hat{E}, \hat{C} = \hat{F}$ $\therefore \triangle ABC \sim \triangle DEF$ |
| SSS<br>(sides in prop.)      |  $\frac{MN}{RS} = \frac{ML}{RT} = \frac{NL}{ST}$ $\therefore \triangle MNL \sim \triangle RST$           |



- If  $O$  is the centre and  $OM \perp AB$ , then  $AM = MB$ .
- If  $O$  is the centre and  $AM = MB$ , then  $\hat{A}MO = \hat{B}MO = 90^\circ$ .
- If  $AM = MB$  and  $OM \perp AB$ , then  $\Rightarrow MO$  passes through centre  $O$ .

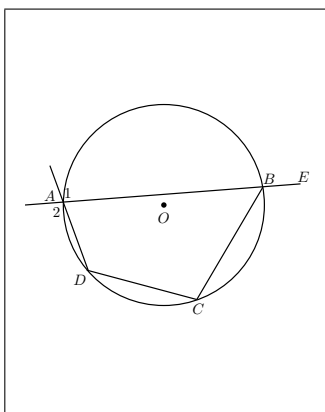


If an arc subtends an angle at the centre of a circle and at the circumference, then the angle at the centre is twice the size of the angle at the circumference.



Angles at the circumference subtended by arcs of equal length (or by the same arc) are equal.

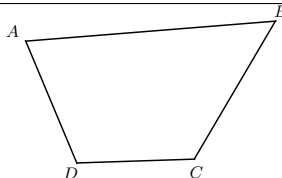
## Cyclic quadrilaterals

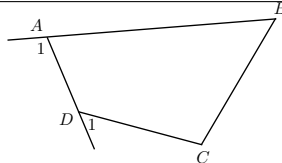


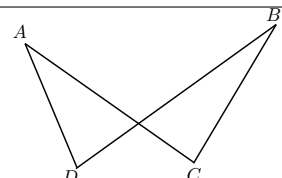
If the four sides of a quadrilateral  $ABCD$  are the chords of a circle with centre  $O$ , then:

- $\hat{A} + \hat{C} = 180^\circ$   
Reason: (opp.  $\angle$ s cyclic quad. supp.)
- $\hat{B} + \hat{D} = 180^\circ$   
Reason: (opp.  $\angle$ s cyclic quad. supp.)
- $\hat{EBC} = \hat{D}$   
Reason: (ext.  $\angle$  cyclic quad. = int. opp  $\angle$ )
- $\hat{A}_1 = \hat{A}_2 = \hat{C}$   
Reason: (vert. opp.  $\angle$ s, ext.  $\angle$  cyclic quad.)

## Proving a quadrilateral is cyclic:

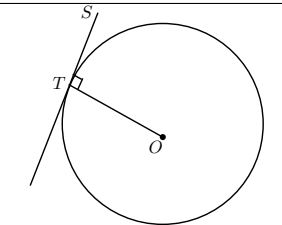
|                                                                                   |                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>If <math>\hat{A} + \hat{C} = 180^\circ</math> or <math>\hat{B} + \hat{D} = 180^\circ</math>, then <math>ABCD</math> is a cyclic quadrilateral.</p> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

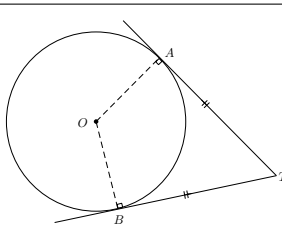
|                                                                                   |                                                                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|  | <p>If <math>\hat{A}_1 = \hat{C}</math> or <math>\hat{D}_1 = \hat{B}</math>, then <math>ABCD</math> is a cyclic quadrilateral.</p> |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|

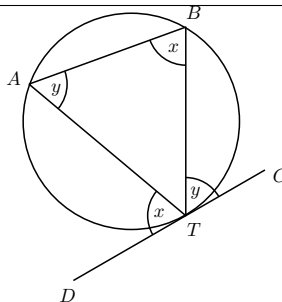
|                                                                                   |                                                                                                                               |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|  | <p>If <math>\hat{A} = \hat{B}</math> or <math>\hat{C} = \hat{D}</math>, then <math>ABCD</math> is a cyclic quadrilateral.</p> |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|

## Tangents to a circle

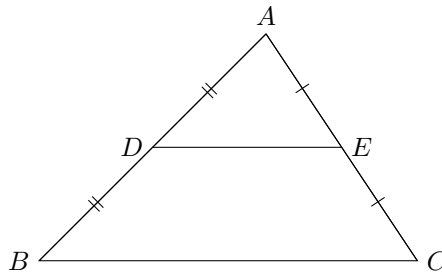
EMCJ6

|                                                                                    |                                                                                                                            |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|  | <p>A tangent is perpendicular to the radius (<math>OT \perp ST</math>), drawn to the point of contact with the circle.</p> |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|

|                                                                                     |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>If <math>AT</math> and <math>BT</math> are tangents to a circle with centre <math>O</math>, then:</p> <ul style="list-style-type: none"> <li>• <math>OA \perp AT</math> (tangent <math>\perp</math> radius)</li> <li>• <math>OB \perp BT</math> (tangent <math>\perp</math> radius)</li> <li>• <math>TA = TB</math> (tangents from same point are equal)</li> </ul> |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                     |                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• If <math>DC</math> is a tangent, then <math>D\hat{T}A = T\hat{B}A</math> and <math>C\hat{T}B = T\hat{A}B</math>.</li> <li>• If <math>D\hat{T}A = T\hat{B}A</math> or <math>C\hat{T}B = T\hat{A}B</math>, then <math>DC</math> is a tangent touching at <math>T</math>.</li> </ul> |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

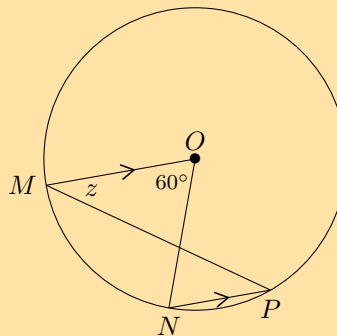
The line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.



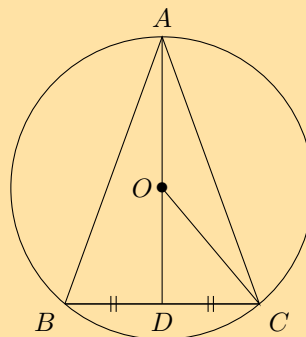
Given:  $AD = DB$  and  $AE = EC$ , we can conclude that  $DE \parallel BC$  and  $DE = \frac{1}{2}BC$ .

### Exercise 8 – 1: Revision

1.  $MO \parallel NP$  in a circle with centre  $O$ .  $\hat{M}ON = 60^\circ$  and  $\hat{OMP} = z$ . Calculate the value of  $z$ , giving reasons.



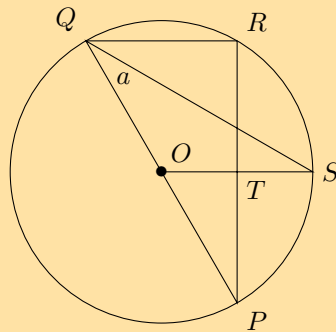
2.  $O$  is the centre of the circle with  $OC = 5$  cm and chord  $BC = 8$  cm.



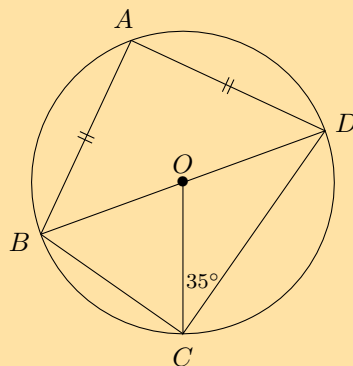
Determine the lengths of:

- a)  $OD$
- b)  $AD$
- c)  $AB$

3.  $PQ$  is a diameter of the circle with centre  $O$ .  $SQ$  bisects  $\angle PQR$  and  $\angle PQS = a$ .

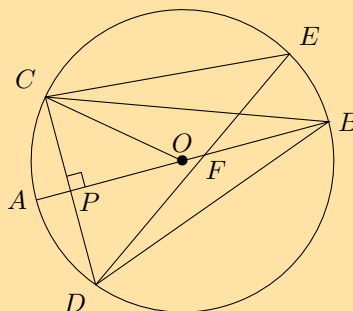


- Write down two other angles that are also equal to  $a$ .
  - Calculate  $\angle POS$  in terms of  $a$ , giving reasons.
  - Prove that  $OS$  is a perpendicular bisector of  $PR$ .
4.  $BD$  is a diameter of the circle with centre  $O$ .  $AB = AD$  and  $\angle OCD = 35^\circ$ .



Calculate the value of the following angles, giving reasons:

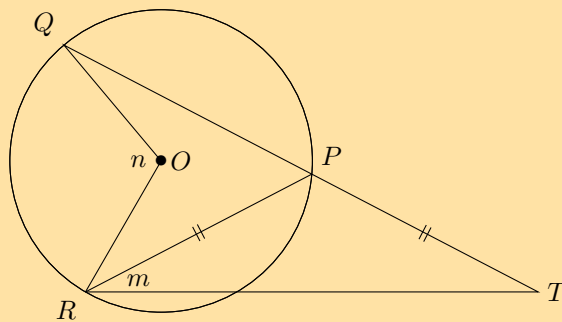
- $\angle ODC$
  - $\angle COD$
  - $\angle CBD$
  - $\angle BAD$
  - $\angle ADB$
5.  $O$  is the centre of the circle with diameter  $AB$ .  $CD \perp AB$  at  $P$  and chord  $DE$  intersects  $AB$  at  $F$ .



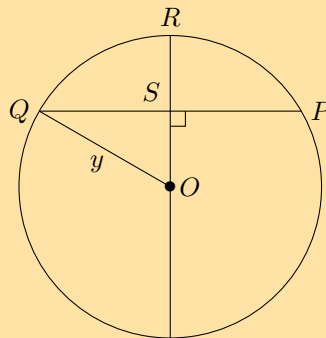
Prove the following:

- $\angle CBP = \angle DBP$
- $\angle CED = 2\angle CBA$
- $\angle ABD = \frac{1}{2}\angle COA$

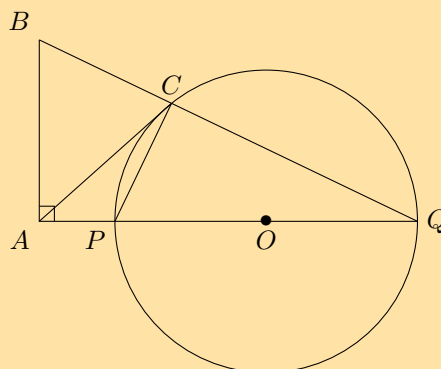
6.  $QP$  in the circle with centre  $O$  is extended to  $T$  so that  $PR = PT$ . Express  $m$  in terms of  $n$ .



7. In the circle with centre  $O$ ,  $OR \perp QP$ ,  $QP = 30$  mm and  $RS = 9$  mm. Determine the length of  $y$ .



8.  $PQ$  is a diameter of the circle with centre  $O$ .  $QP$  is extended to  $A$  and  $AC$  is a tangent to the circle.  $BA \perp AQ$  and  $BCQ$  is a straight line.

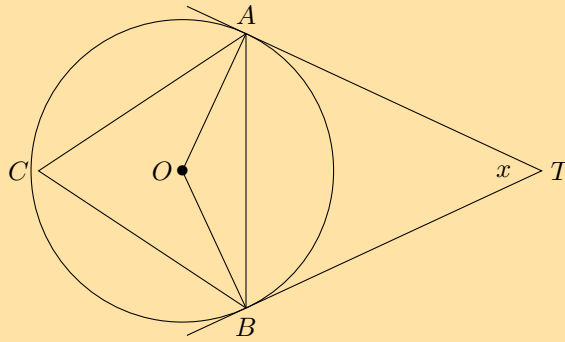


Prove the following:

- $\angle PCQ = \angle BAP$
- $BAPC$  is a cyclic quadrilateral
- $AB = AC$



9.  $TA$  and  $TB$  are tangents to the circle with centre  $O$ .  $C$  is a point on the circumference and  $\hat{ATB} = x$ .



Express the following in terms of  $x$ , giving reasons:

a)  $\hat{ABT}$

b)  $\hat{OBA}$

c)  $\hat{C}$

10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 2982   2. 2983   3. 2984   4. 2985   5. 2986   6. 2987  
7. 2988   8. 2989   9. 298B



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## 8.2 Ratio and proportion

EMCJ8

### Ratio

A ratio describes the relationship between two quantities which have the same units. We can use ratios to compare weights, heights, lengths, currencies, etc. A ratio is a comparison between two quantities of the same kind and has no units.

Example: if the length of a rectangle is 20 cm and the width is 60 cm, then we can express the ratio between the length and width of the rectangle as:

$$\begin{array}{lll} \text{length to width} = 20 \text{ to } 60 & \text{Or} & \frac{\text{length}}{\text{width}} = \frac{20}{60} \\ & & = \frac{1}{3} \end{array} \quad \text{Or} \quad \begin{array}{l} \text{length : width} = 20 : 60 \\ = 1 : 3 \end{array}$$

- The ratio of  $\frac{1}{3}$  describes the length of the rectangle **relative** to its width.
- A ratio written as a fraction is usually given in its simplest form.
- A ratio gives no indication of actual length. For example,

$$\frac{\text{length}}{\text{width}} = \frac{50 \text{ cm}}{150 \text{ cm}} \text{ also gives a ratio of } \frac{1}{3}$$

$$\text{And } \frac{\text{length}}{\text{width}} = \frac{0,8 \text{ m}}{2,4 \text{ m}} \text{ also gives a ratio of } \frac{1}{3}$$

- Do not convert a ratio to a decimal (even though  $\frac{1}{3}$  and  $0,3$  have the same numerical value).

## Proportion

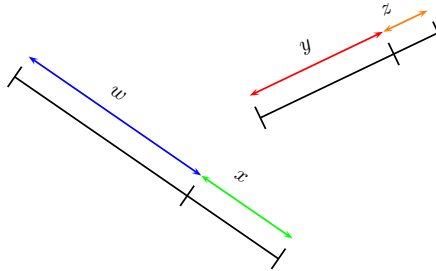
### Investigation: Predicting heights

A record of heights is given.

If the ratio  $\frac{\text{height of a person at two years old}}{\text{height of a person as an adult}}$  is 1 to 2, complete the table below:

| Name    | Height at two years | Height as an adult |
|---------|---------------------|--------------------|
| Hendrik | 84 cm               |                    |
| Kagiso  |                     | 162 cm             |
| Linda   | 86 cm               |                    |
| Mandisa | 0,87 m              |                    |
| Prashna |                     | 1 m 64 cm          |

Consider the diagram below:



If two or more ratios are equal to each other, then we say that they are in the same proportion. Proportionality describes the equality of ratios.

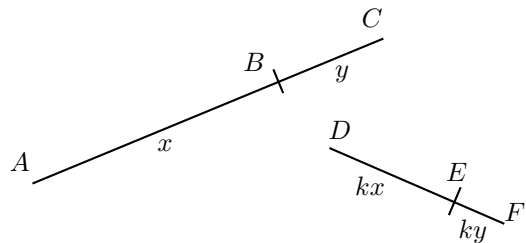
If  $\frac{w}{x} = \frac{y}{z}$ , then  $w$  and  $x$  are in the same proportion as  $y$  and  $z$ .

$$1. \quad wz = xy$$

$$2. \quad \frac{x}{w} = \frac{z}{y}$$

$$3. \quad \frac{w}{y} = \frac{x}{z}$$

$$4. \quad \frac{y}{w} = \frac{z}{x}$$



Given

$$\frac{AB}{BC} = \frac{x}{y} = \frac{kx}{ky} = \frac{DE}{EF}$$

The line segments  $AB$  and  $BC$  are in the same proportion as  $DE$  and  $EF$ . The following statements are also true:

| Proportion                      | Reciprocal proportion           | Cross multiplication        |
|---------------------------------|---------------------------------|-----------------------------|
| $\frac{AB}{BC} = \frac{DE}{FE}$ | $\frac{BC}{AB} = \frac{FE}{DE}$ | $AB \cdot FE = BC \cdot DE$ |
| $\frac{AB}{AC} = \frac{DE}{DF}$ | $\frac{AC}{AB} = \frac{DF}{DE}$ | $AB \cdot DF = AC \cdot DE$ |
| $\frac{BC}{AC} = \frac{EF}{DF}$ | $\frac{AC}{BC} = \frac{DF}{EF}$ | $BC \cdot DF = AC \cdot EF$ |

We can also substitute  $x$ ,  $y$ ,  $kx$ , and  $ky$  to show algebraically that the statements are true.

For example,

$$\begin{aligned} BC \cdot DF &= y \times (kx + ky) \\ &= ky(x + y) \\ \text{And } AC \cdot EF &= (x + y) \times ky \\ &= ky(x + y) \\ \therefore BC \cdot DF &= AC \cdot EF \end{aligned}$$

## Exercise 8 – 2: Ratio and proportion

1. Solve for  $p$ :

a)  $\frac{8}{40} = \frac{p}{25}$

b)  $\frac{6}{9} = \frac{29+p}{54}$

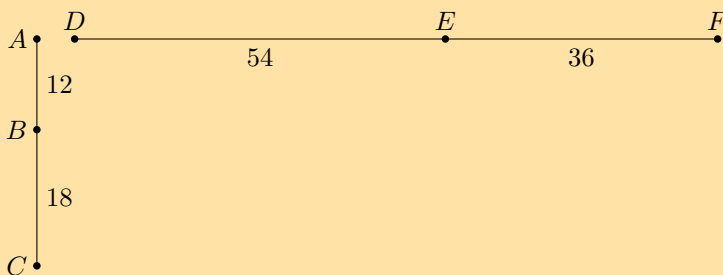
c)  $\frac{3}{1 + \frac{p}{4}} = \frac{4}{p+1}$

d)  $\frac{14}{100-p} = \frac{49}{343}$

2. A packet of 160 sweets contains red, blue and yellow sweets in the ratio of 3 : 2 : 3 respectively. Determine how many sweets of each colour there are in the packet.

3. A mixture contains 2 parts of substance  $A$  for every 5 parts of substance  $B$ . If the total weight of the mixture is 50 kg, determine how much of substance  $B$  is in the mixture (correct to 2 decimal places).

4. Given the diagram below.



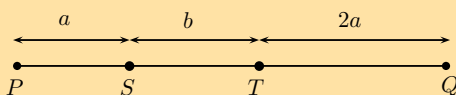
Show that:

a)  $\frac{AB}{BC} = \frac{FE}{ED}$

b)  $\frac{AC}{BC} = \frac{FD}{EF}$

c)  $AB \cdot DF = AC \cdot FE$

5. Consider the line segment shown below.



Express the following in terms of  $a$  and  $b$ :

a)  $PT : ST$

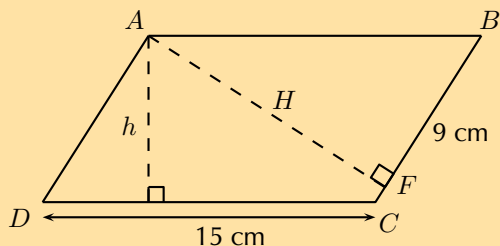
b)  $\frac{PS}{TQ}$

c)  $\frac{SQ}{PQ}$

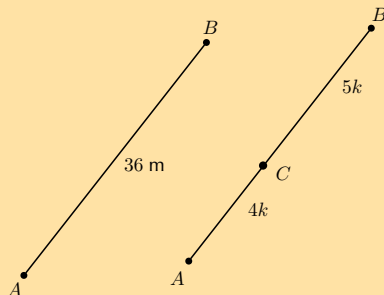
d)  $QT : TS$

6.  $ABCD$  is a parallelogram with  $DC = 15$  cm,  $h = 8$  cm and  $BF = 9$  cm.

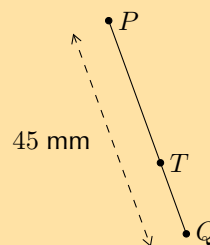
Calculate the ratio  $\frac{\text{area } ABF}{\text{area } ABCD}$ .



7.  $AB = 36$  m and  $C$  divides  $AB$  in the ratio  $4 : 5$ . Determine  $AC$  and  $CB$ .



8. If  $PQ = 45$  mm and the ratio of  $TQ : PQ$  is  $2 : 3$ , calculate  $PT$  and  $TQ$ .

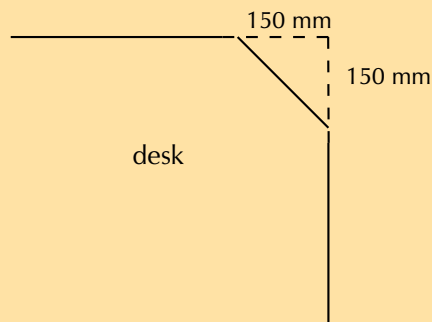


9. Luke's biology notebook is 30 cm long and 20 cm wide. The dimensions of his desk are in the same proportion as the dimensions of his notebook.

a) If the desk is 90 cm wide, calculate the area of the top of the desk.

b) Luke covers each corner of his desk with an isosceles triangle of cardboard, as shown in the diagram on the right.

Calculate the new perimeter and area of the visible part of the top of his desk.



c) Use this new area to calculate the dimensions of a square desk with the same desk top area.

10. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

- 1a. 298C   1b. 298D   1c. 298F   1d. 298G   2. 298H   3. 298J  
4a. 298K   4b. 298M   4c. 298N   5a. 298P   5b. 298Q   5c. 298R  
5d. 298S   6. 298T   7. 298V   8. 298W   9. 298X



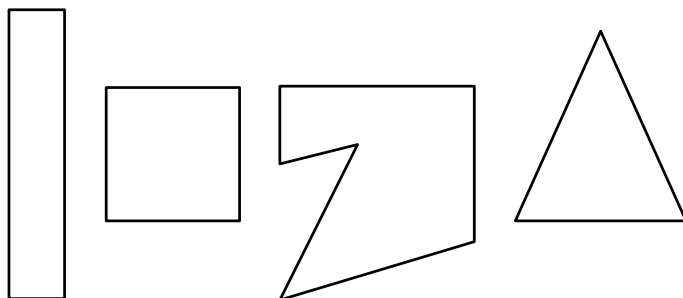
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**DEFINITION:** *Proportionality in polygons*

A plane, closed shape consisting of three or more line segments.



In previous grades we studied the properties of the following polygons:

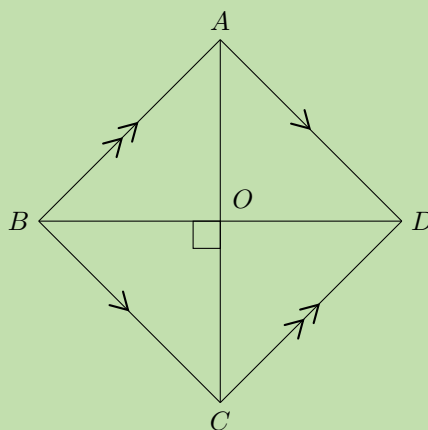
|               |  |                                  |
|---------------|--|----------------------------------|
| Triangle      |  | Area = $\frac{1}{2}b \times h$   |
| Parallelogram |  | Area = $b \times h$              |
| Rectangle     |  | Area = $b \times h$              |
| Rhombus       |  | Area = $\frac{1}{2}AC \times BD$ |

|           |  |                                             |
|-----------|--|---------------------------------------------|
| Square    |  | $\text{Area} = s^2$                         |
| Trapezium |  | $\text{Area} = \frac{1}{2}(a + b) \times h$ |
| Kite      |  | $\text{Area} = \frac{1}{2}(AC \times DB)$   |

### Worked example 1: Properties of polygons

#### QUESTION

$ABCD$  is a rhombus with  $BD = 12$  cm and  $AB : BD = 3 : 4$ .



Calculate the following (correct to two decimal places) and provide reasons:

1. Length of  $AB$
2. Length of  $AO$
3. Area of  $ABCD$

## SOLUTION

**Step 1: Use the ratio to determine the length of AB**

$$\begin{aligned}AB : BD &= 3 : 4 \\ \therefore \frac{AB}{BD} &= \frac{3}{4} \\ \frac{AB}{12} &= \frac{3}{4} \\ AB &= 12 \times \frac{3}{4} \\ &= 9 \text{ cm}\end{aligned}$$

**Step 2: Calculate the length of AO**

We use the properties of a rhombus and the theorem of Pythagoras to find  $AO$ .

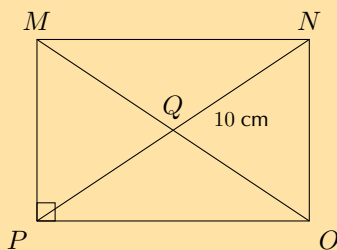
$$\begin{aligned}BD &= 12 \text{ cm} \\ BO &= 6 \text{ cm} && \text{(diagonals bisect each other)} \\ \text{In } \triangle ABO, \quad \hat{AOB} &= 90^\circ && \text{(diagonals intersect at } \perp) \\ AO^2 &= AB^2 - BO^2 && \text{(Pythagoras)} \\ &= 9^2 - 6^2 \\ \therefore AO &= \sqrt{45} \\ &= 6,71 \text{ cm}\end{aligned}$$

**Step 3: Determine the area of rhombus ABCD**

$$\begin{aligned}\text{Area } ABCD &= \frac{1}{2} AC \times BD \\ &= \frac{1}{2} (2 \times \sqrt{45})(12) \\ &= 80,50 \text{ cm}^2\end{aligned}$$

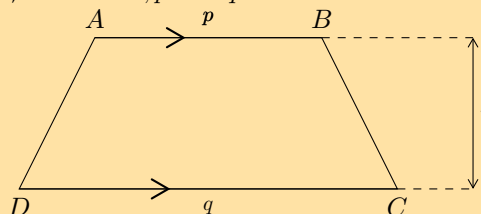
## Exercise 8 – 3: Proportionality of polygons

1.  $MNOP$  is a rectangle with  $MN : NO = 5 : 3$  and  $QN = 10$  cm.

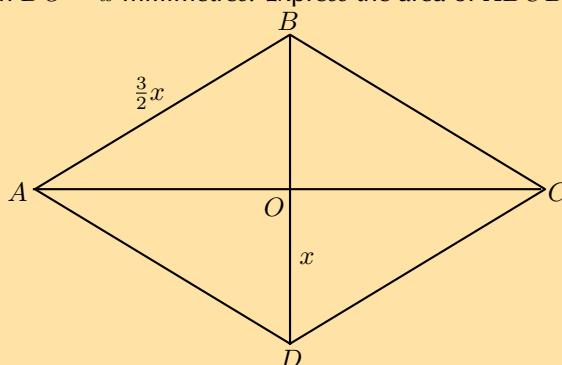


- Calculate  $MN$  (correct to 2 decimal places).
- Calculate the area of  $\triangle OPQ$  (correct to 2 decimal places).

2. Consider the trapezium  $ABCD$  shown below. If  $t : p : q = 2 : 3 : 5$  and area  $ABCD = 288 \text{ cm}^2$ , calculate  $t, p$  and  $q$ .

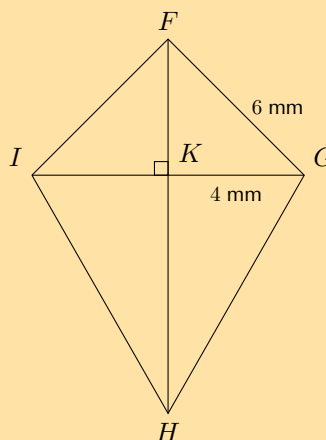


3.  $ABCD$  is a rhombus with sides of length  $\frac{3}{2}x$  millimetres. The diagonals intersect at  $O$  and length  $DO = x$  millimetres. Express the area of  $ABCD$  in terms of  $x$ .

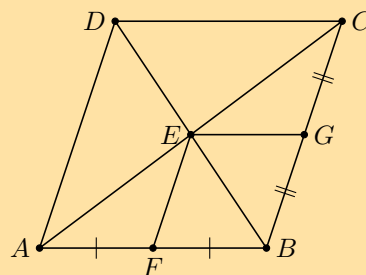


4. In the diagram below,  $FGHI$  is a kite with  $FG = 6 \text{ mm}$ ,  $GK = 4 \text{ mm}$  and  $\frac{GH}{FI} = \frac{5}{2}$ .

- Determine  $FH$  (correct to the nearest integer).
- Calculate area  $FGHI$ .



5.  $ABCD$  is a rhombus.  $F$  is the mid-point of  $AB$  and  $G$  is the mid-point of  $CB$ . Prove that  $EFG$  is also a rhombus.



6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.  
Check answers online with the exercise code below or click on 'show me the answer'.

1. 298Y   2. 298Z   3. 2992   4. 2993   5. 2994



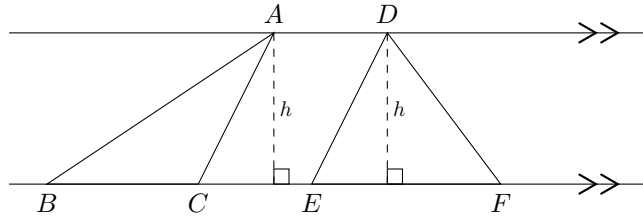
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1. In the diagram below,  $\triangle ABC$  and  $\triangle DEF$  have the same height ( $h$ ) since both triangles are between the same parallel lines.



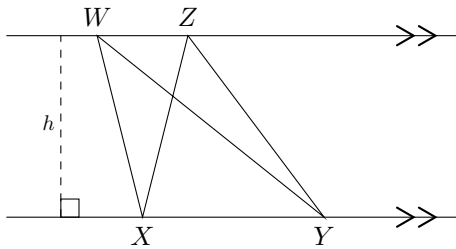
$$\text{Area } \triangle ABC = \frac{1}{2} BC \times h$$

$$\text{Area } \triangle DEF = \frac{1}{2} EF \times h \quad \text{and} \quad \frac{\text{Area } \triangle ABC}{\text{Area } \triangle DEF} = \frac{\frac{1}{2} BC \times h}{\frac{1}{2} EF \times h}$$

$$\text{Therefore } \frac{\text{Area } \triangle ABC}{\text{Area } \triangle DEF} = \frac{BC}{EF}$$

Triangles with equal heights have areas which are proportional to their bases.

2.  $\triangle WXY$  and  $\triangle ZXY$  have the same base ( $XY$ ) and the same height ( $h$ ) since both triangles lie between the same parallel lines.



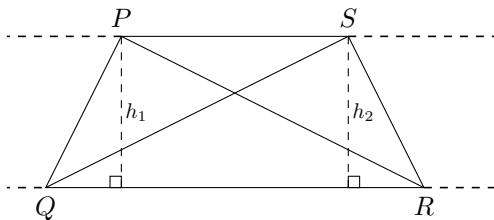
$$\text{Area } \triangle WXY = \frac{1}{2} XY \times h$$

$$\text{Area } \triangle ZXY = \frac{1}{2} XY \times h$$

$$\therefore \text{Area } \triangle WXY = \text{Area } \triangle ZXY$$

Triangles with equal bases and between the same parallel lines are equal in area.

3.  $\triangle PQR$  and  $\triangle SQR$  have the same base ( $QR$ ) and are equal in area.



$$\text{Area } \triangle PQR = \text{Area } \triangle SQR$$

$$\frac{1}{2} QR \times h_1 = \frac{1}{2} QR \times h_2$$

$$\therefore h_1 = h_2$$

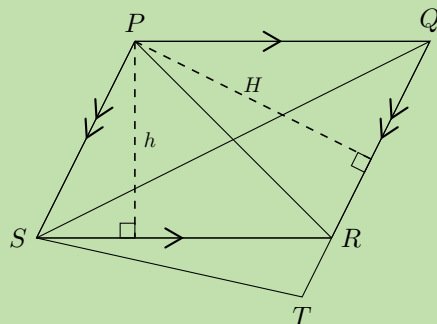
$$\therefore PS \parallel QR$$

Triangles on the same side of the same base and equal in area, lie between parallel lines.

## Worked example 2: Proportionality of triangles

### QUESTION

Given parallelogram  $PQRS$  with  $QR$  produced to  $T$ .  $RS = 45$  cm,  $QR = 30$  cm and  $h = 10$  cm.



1. Calculate  $H$ .
2. If  $TR : TQ = 1 : 4$ , show that  $\frac{\text{Area } \triangle STR}{\text{Area } \triangle PRQ} = \frac{1}{3}$ .

### SOLUTION

#### Step 1: Determine the length of $H$

We use the formula for area of a parallelogram to calculate  $H$ .

$$\begin{aligned}\text{Area } PQRS &= SR \times h \\ &= 45 \times 10 \\ &= 450 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area } PQRS &= QR \times H \\ 450 &= 30 \times H \\ \therefore H &= 15 \text{ cm}\end{aligned}$$

#### Step 2: Use proportionality to show that $\frac{\text{Area } \triangle STR}{\text{Area } \triangle PRQ} = \frac{1}{3}$

We are given the ratio  $TR : TQ = 1 : 4$ .

$$\begin{aligned}\frac{TR}{TQ} &= \frac{1}{4} \\ \text{So then } \frac{RQ}{TQ} &= \frac{3}{4} \\ \text{And } \frac{TR}{RQ} &= \frac{TR}{TQ} \times \frac{TQ}{RQ} \\ &= \frac{1}{4} \times \frac{4}{3} \\ &= \frac{1}{3}\end{aligned}$$

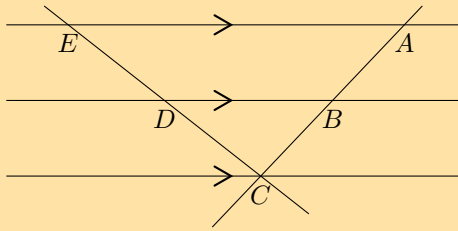
$$\text{Area } \triangle STR = \frac{1}{2} TR \times H \quad (PS \parallel QT, \text{ equal heights})$$

$$\text{Area } \triangle PRQ = \frac{1}{2} RQ \times H$$

$$\begin{aligned}\therefore \frac{\text{Area } \triangle STR}{\text{Area } \triangle PRQ} &= \frac{\frac{1}{2} TR \times H}{\frac{1}{2} RQ \times H} \\ &= \frac{TR}{RQ} \\ &= \frac{1}{3}\end{aligned}$$

## Exercise 8 – 4: Proportionality of triangles

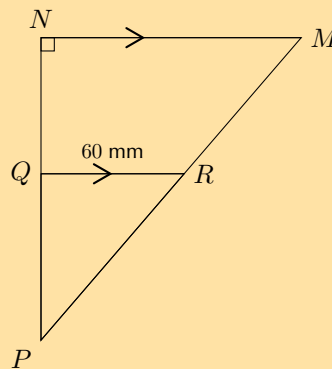
1. The diagram below shows three parallel lines cut by two transversals  $EC$  and  $AC$  such that  $ED : DC = 4 : 6$ .



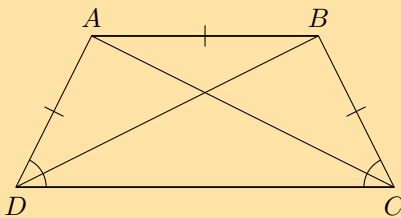
Determine:

- $\frac{BC}{AB}$
- $AB : AC$
- The lengths of  $AC$  and  $ED$ , if it is given that  $AB = 12$  mm.

2. In right-angled  $\triangle MNP$ ,  $QR$  is drawn parallel to  $NM$ , with  $R$  the mid-point of  $MP$ .  $NP = 16$  cm and  $RQ = 60$  mm. Determine  $QP$  and  $RP$ .

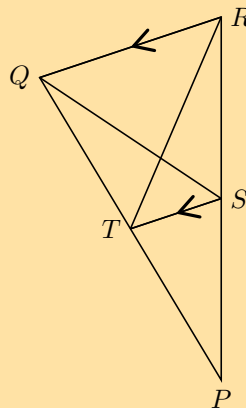


3. Given trapezium  $ABCD$  with  $DA = AB = BC$  and  $\hat{ADC} = \hat{BCD}$ .



- Prove that  $BD$  bisects  $\hat{D}$ .
- Prove that the two diagonals are equal in length.
- If  $DC : AB = 5 : 4$ , show that  $\text{area } ABCD = 2,25 \times \text{area } \triangle ABC$ .

4. In the diagram to the right,  $\triangle PQR$  is given with  $QR \parallel TS$ . Show that  $\text{area } \triangle PQS = \text{area } \triangle PRT$ .

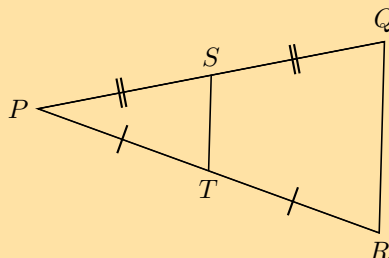


5. In Grade 10 we proved the mid-point theorem using congruent triangles.

a) Complete the following statement of the mid-point theorem:

"The line joining ..... of a triangle is ..... to the third side and equal to ....."

b) In  $\triangle PQR$ ,  $T$  and  $S$  are the mid-points of  $PR$  and  $PQ$  respectively. Prove  $TS \parallel RQ$ .



c) Write down the converse of the mid-point theorem.

6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 2995    2. 2996    3. 2997    4. 2998    5. 2999



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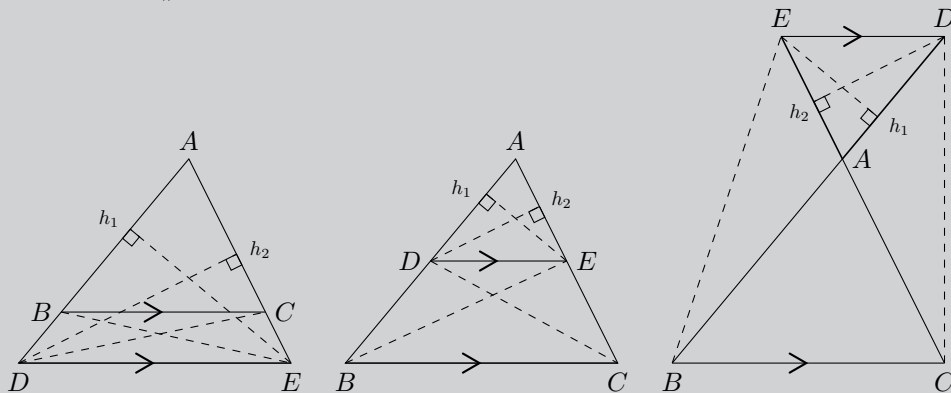
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### Theorem: Proportion theorem

#### STATEMENT

A line drawn parallel to one side of a triangle divides the other two sides of the triangle in the same proportion.

(Reason: line  $\parallel$  to one side  $\triangle$ )



**Given:**

$\triangle ABC$  with line  $DE \parallel BC$

**Required to prove:**

$$\frac{AD}{DB} = \frac{AE}{EC}$$

### PROOF

Draw  $h_1$  from  $E$  perpendicular to  $AD$ , and  $h_2$  from  $D$  perpendicular to  $AE$ .

Draw  $BE$  and  $CD$ .

$$\frac{\text{Area } \triangle ADE}{\text{Area } \triangle BDE} = \frac{\frac{1}{2}AD \cdot h_1}{\frac{1}{2}DB \cdot h_1} = \frac{AD}{DB}$$

$$\frac{\text{Area } \triangle ADE}{\text{Area } \triangle CED} = \frac{\frac{1}{2}AE \cdot h_2}{\frac{1}{2}EC \cdot h_2} = \frac{AE}{EC}$$

but  $\text{Area } \triangle BDE = \text{Area } \triangle CED$  (equal base and height)

$$\therefore \frac{\text{Area } \triangle ADE}{\text{Area } \triangle BDE} = \frac{\text{Area } \triangle ADE}{\text{Area } \triangle CED}$$

$$\therefore \frac{AD}{DB} = \frac{AE}{EC}$$

Similarly, we use the same method to show that:

$$\frac{AD}{AB} = \frac{AE}{AC} \quad \text{and} \quad \frac{AB}{BD} = \frac{AC}{CE}$$

### Converse: proportion theorem

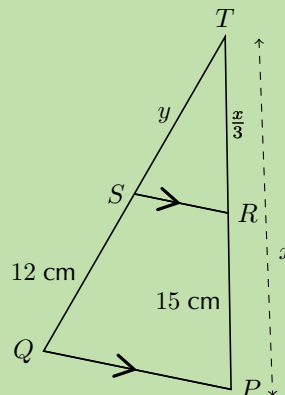
If a line divides two sides of a triangle in the same proportion, then the line is parallel to the third side.

(Reason: line divides sides in prop.)

### Worked example 3: Proportion theorem

#### QUESTION

In  $\triangle TQP$ ,  $SR \parallel QP$ ,  $SQ = 12$  cm and  $RP = 15$  cm. If  $TR = \frac{x}{3}$ ,  $TP = x$  and  $TS = y$ , determine the values of  $x$  and  $y$ , giving reasons.



### SOLUTION

**Step 1: Use the proportion theorem to determine the values of  $x$  and  $y$**

Consider  $TP$ :

$$\begin{aligned} RP &= x - \frac{1}{3}x \\ &= \frac{2}{3}x \end{aligned}$$

$$\begin{aligned} \therefore 15 &= \frac{2}{3}x \\ x &= 22,5 \text{ cm} \end{aligned}$$

$$\begin{aligned} TR &= \frac{1}{3}(22,5) \\ &= 7,5 \text{ cm} \end{aligned}$$

Consider  $TQ$ :

$$\frac{TS}{SQ} = \frac{TR}{RP} \quad (\text{line} \parallel \text{one side of } \triangle)$$

$$\frac{y}{12} = \frac{7,5}{15}$$

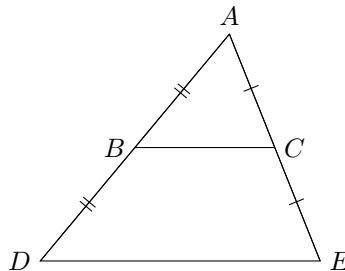
$$\therefore y = 6 \text{ cm}$$

### Special case: the mid-point theorem

A corollary of the proportion theorem is the mid-point theorem: the line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.

If  $AB = BD$  and  $AC = CE$ , then  $BC \parallel DE$  and  $BC = \frac{1}{2}DE$ .

We also know that  $\frac{AC}{CE} = \frac{AB}{BD}$ .



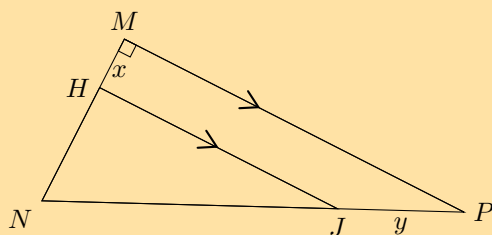
### Converse: the mid-point theorem

The line drawn from the mid-point of one side of a triangle parallel to another side, bisects the third side of the triangle.

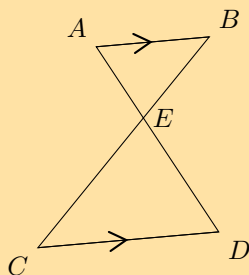
If  $AB = BD$  and  $BC \parallel DE$ , then  $AC = CE$ .

## Exercise 8 – 5: Proportion theorem

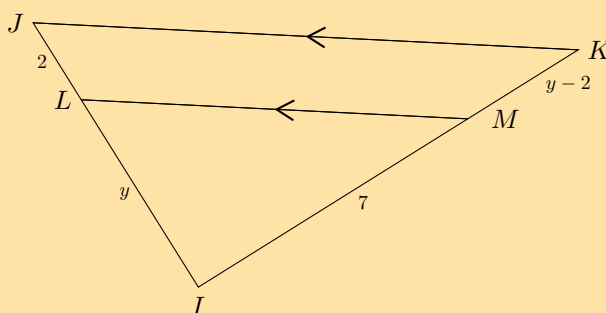
1. In  $\triangle MNP$ ,  $\hat{M} = 90^\circ$  and  $HJ \parallel MP$ .  
 $HN : MH = 3 : 1$ ,  $HM = x$  and  $JP = y$ .



- a) Calculate  $JP : NP$ .  
 b) Calculate  $\frac{\text{area } \triangle HNJ}{\text{area } \triangle MNP}$ .
2. Use the given diagram to prove the Proportion Theorem.

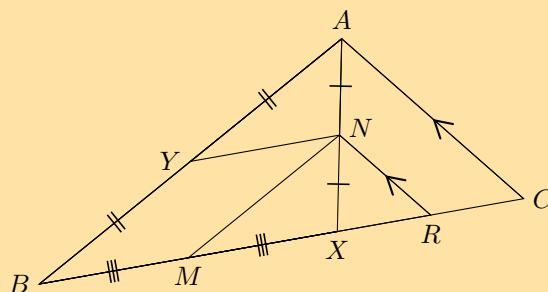


3. In the diagram below,  $JL = 2$ ,  $LI = y$ ,  $IM = 7$  and  $MK = y - 2$ .  
 If  $LM \parallel JK$ , calculate  $y$  (correct to two decimal places).



4. Write down the converse of the proportion theorem and illustrate with a diagram.

5. In  $\triangle ABC$ ,  $X$  is a point on  $BC$ .  $N$  is the mid-point of  $AX$ ,  $Y$  is the mid-point of  $AB$  and  $M$  is the mid-point of  $BX$ .



- Prove that  $YBMN$  is a parallelogram.
- Prove that  $MR = \frac{1}{2}BC$ .

6. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 299B   2. 299C   3. 299D   4. 299F   5. 299G



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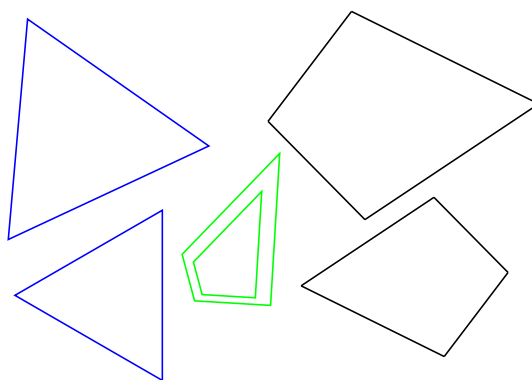
## 8.5 Similarity

EMCJD

### Similar polygons

EMCJF

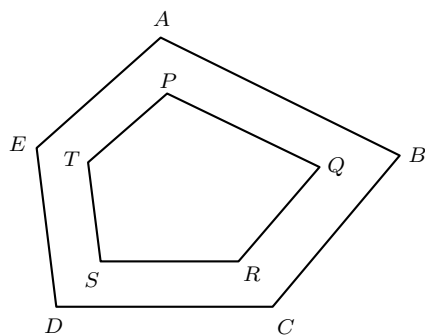
Polygons are similar if they are the same shape but differ in size. In other words, one polygon is an enlargement of the other. Congruent polygons are also similar since they are the same shape and the same size, only their position or orientation is different.



Two polygons with the same number of sides are similar when:

- All pairs of corresponding angles are equal, **and**
- All pairs of corresponding sides are in the same proportion.





In the diagram above, polygon  $ABCDE$  is similar to polygon  $PQRST$  if:

Condition 1:

$$\hat{A} = \hat{P}, \quad \hat{B} = \hat{Q}, \quad \hat{C} = \hat{R}, \quad \hat{D} = \hat{S}, \quad \hat{E} = \hat{T}$$

**AND**

Condition 2:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CD}{RS} = \frac{DE}{ST} = \frac{EA}{TP}$$

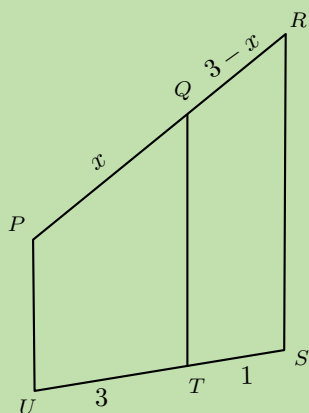
**Important:**

- Both conditions must be true for two polygons to be similar.
- If we are given two polygons that are similar, then we know that both conditions are true.

#### Worked example 4: Similar polygons

##### QUESTION

Polygons  $PQTU$  and  $PRSU$  are similar. Determine the value of  $x$ .



## SOLUTION

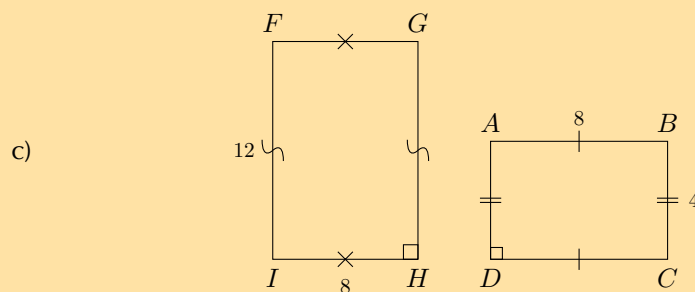
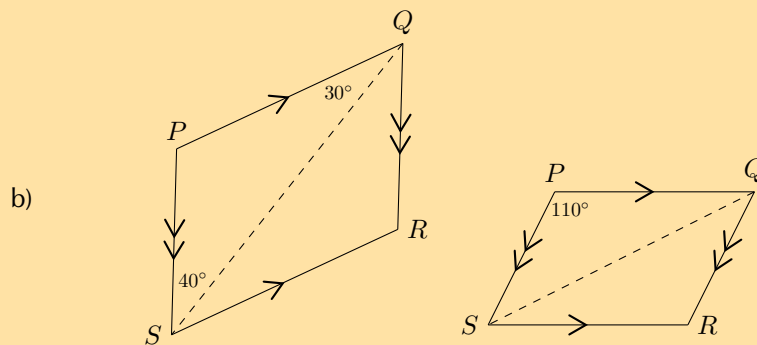
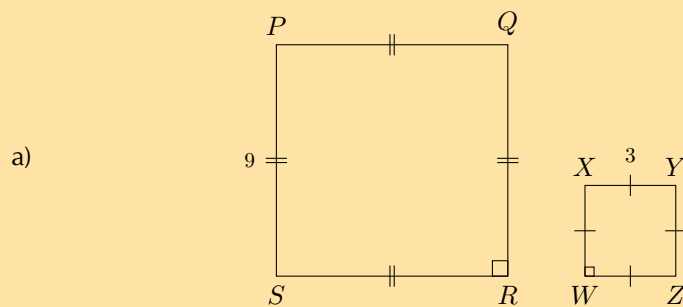
### Step 1: Identify pairs of corresponding sides

Since the two polygons are similar,

$$\begin{aligned}\frac{PQ}{PR} &= \frac{TU}{SU} \\ \therefore \frac{x}{x + (3 - x)} &= \frac{3}{3 + 1} \\ \therefore \frac{x}{3} &= \frac{3}{4} \\ \therefore x &= \frac{9}{4}\end{aligned}$$

## Exercise 8 – 6: Similar polygons

1. Determine whether or not the following polygons are similar, giving reasons.



2. Are the following statements true or false? If false, state reasons or draw an appropriate diagram.

- a) All squares are similar.
- b) All rectangles are similar.
- c) All rhombi are similar.
- d) All congruent polygons are similar.
- e) All similar polygons are congruent.
- f) All congruent triangles are similar.
- g) Isosceles triangles are similar.
- h) Equilateral triangles are similar.

3. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 299H   1b. 299J   1c. 299K   2a. 299M   2b. 299N   2c. 299P  
2d. 299Q   2e. 299R   2f. 299S   2g. 299T   2h. 299V



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## Similarity of triangles

EMCJG

To prove two polygons are similar, we need to show that two conditions are true: (a) all pairs of corresponding angles are equal and (b) all pairs of corresponding sides are in the same proportion.

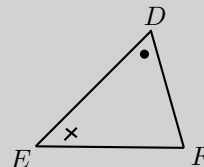
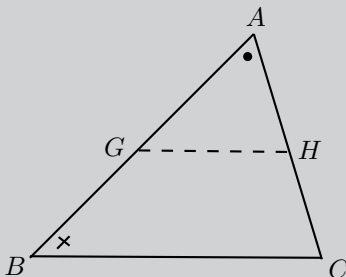
To prove two triangles are similar, we need only show that **one** of the conditions is true. If one of the conditions is true for two triangles, then the other condition is also true.

### Theorem: Equiangular triangles are similar

#### STATEMENT

Equiangular triangles are similar.

(Reason: equiangular  $\triangle$ s)



#### Given:

$\triangle ABC$  and  $\triangle DEF$   
with  $\hat{A} = \hat{D}$ ;  $\hat{B} = \hat{E}$ ;  $\hat{C} = \hat{F}$

#### Required to prove:

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$

## PROOF

**Construction:** mark  $G$  on  $AB$  so that  $AG = DE$ , and mark  $H$  on  $AC$  so that  $AH = DF$ .

In  $\triangle AGH$  and  $\triangle DEF$

$$\begin{aligned}AG &= DE && \text{(by construction)} \\AH &= DF && \text{(by construction)} \\\hat{A} &= \hat{D} && \text{(given)} \\\triangle AGH &\equiv \triangle DEF && \text{(SAS)} \\\therefore \hat{AGH} &= \hat{E} \\ \hat{E} &= \hat{B} && \text{(given)} \\\therefore \hat{AGH} &= \hat{B} \\\therefore GH &\parallel BC && \text{(corresp. } \angle\text{s equal)}\end{aligned}$$

$$\begin{aligned}\frac{AB}{AG} &= \frac{AC}{AH} && \text{(prop. theorem)} \\\therefore \frac{AB}{DE} &= \frac{AC}{DF} && (AG = DE, AH = DF)\end{aligned}$$

Similarly, by constructing  $Q$  on  $CA$  so that  $CQ = FD$ , and mark  $P$  on  $BC$  so that  $CP = FE$ .

$$\begin{aligned}\frac{CA}{FD} &= \frac{CB}{FE} \\\therefore \frac{AB}{DE} &= \frac{AC}{DF} = \frac{BC}{EF} \\\therefore \triangle ABC &\parallel\parallel \triangle DEF\end{aligned}$$

Therefore equiangular triangles are similar.

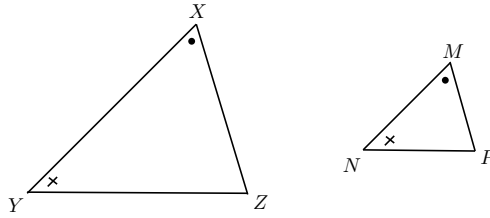
## Notation

- The symbol for congruency is  $\equiv$ .
- The symbol for similarity is  $\parallel\parallel$ .
- Be careful to label similar triangles correctly.  
For example, if  $\hat{P} = \hat{B}$ ,  $\hat{Q} = \hat{A}$  and  $\hat{R} = \hat{C}$ , then  $\triangle PQR \parallel\parallel \triangle BAC$ .  
Do not write  $\triangle PQR \parallel\parallel \triangle ABC$ .
- $\triangle PQR \parallel\parallel \triangle BAC$  also indicates which sides of the triangles are in the same proportion:

$$\frac{PQ}{BA} = \frac{QR}{AC} = \frac{PR}{BC}$$

### Proving equiangular triangles are similar:

The sum of the interior angles of any triangle is  $180^\circ$ . If we know that two pairs of angles are equal, then the remaining angle in each triangle must also be equal. Therefore the two triangles are similar.



$$\hat{X} = 180^\circ - (\hat{Y} + \hat{Z}) \quad (\text{sum of int. } \angle\text{s of } \triangle)$$

$$\hat{M} = 180^\circ - (\hat{N} + \hat{P}) \quad (\text{sum of int. } \angle\text{s of } \triangle)$$

$$\therefore \hat{X} = \hat{M}$$

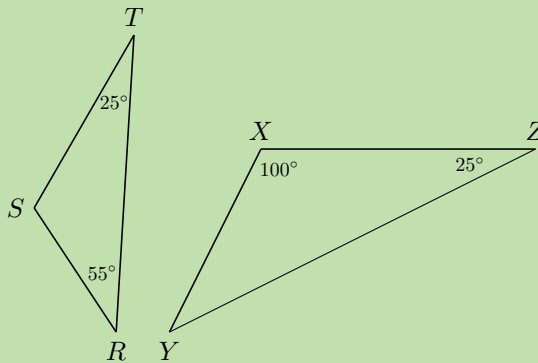
$$\therefore \triangle XYZ \parallel \triangle MNP \quad (\text{equiangular})$$

In other words, to prove that two triangles are equiangular, we need only show that **two** pairs of angles are equal.

### Worked example 5: Similarity of triangles

#### QUESTION

Prove that  $\triangle XYZ$  is similar to  $\triangle SRT$ .



#### SOLUTION

**Step 1: Calculate the unknown angles in each triangle**

$$\begin{aligned} \text{In } \triangle XYZ : \quad \hat{Y} &= 180^\circ - (100^\circ + 25^\circ) && (\text{sum of int. } \angle\text{s of } \triangle) \\ &= 55^\circ \end{aligned}$$

$$\begin{aligned} \text{In } \triangle SRT : \quad \hat{S} &= 180^\circ - (55^\circ + 25^\circ) && (\text{sum of int. } \angle\text{s of } \triangle) \\ &= 100^\circ \end{aligned}$$

**Step 2: Prove that the two triangles are similar**

In  $\triangle XYZ$  and  $\triangle SRT$  :

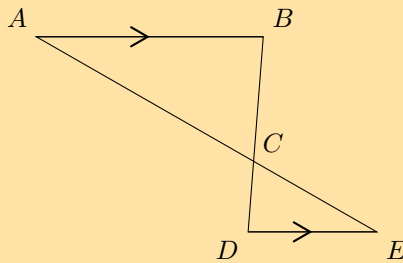
$$\hat{X} = \hat{S} = 100^\circ \quad (\text{proved})$$

$$\hat{Z} = \hat{T} = 25^\circ \quad (\text{given})$$

$$\therefore \triangle XYZ \parallel \triangle SRT \quad (\text{AAA})$$

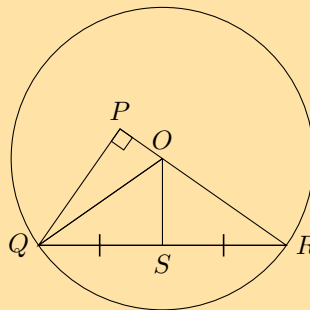
**Exercise 8 – 7: Similarity of triangles**

1. In the diagram below,  $AB \parallel DE$ .



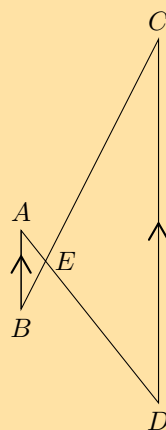
- a) Prove  $\triangle ABC \parallel \triangle EDC$ .  
b) If  $\frac{AC}{AE} = \frac{5}{7}$  and  $AB = 4$  cm, calculate the length of  $DE$  (correct to one decimal place).

2. In circle  $O$ ,  $RP \perp PQ$ .



- a) Prove  $\triangle PRQ \parallel \triangle SRO$ .  
b) Prove  $\frac{OR}{SR} = \frac{QR}{PR}$ .  
c) If  $SR = 18$  mm and  $QP = 20$  mm, calculate the radius of circle  $O$  (correct to one decimal place).

3. Given the opposite figure with the following lengths, find  $AE$ ,  $EC$  and  $BE$ .  
 $BC = 15$  cm,  $AB = 4$  cm,  $CD = 18$  cm and  $ED = 9$  cm.



4. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 299W 2. 299X 3. 299Y



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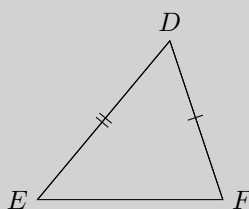
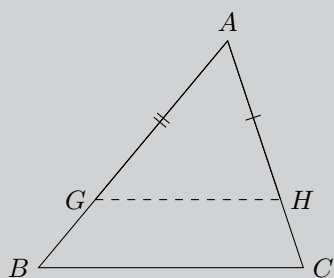
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### Theorem: Triangles with sides in proportion are similar

#### STATEMENT

If the corresponding sides of two triangles are in proportion, then the two triangles are similar.

(Reason: sides of  $\triangle$ s in prop.)



**Given:**

$\triangle ABC$  and  $\triangle DEF$  with  $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$

**Required to prove:**

$\hat{A} = \hat{D}$ ,  $\hat{B} = \hat{E}$  and  $\hat{C} = \hat{F}$

### PROOF

**Construction:** draw  $GH$  such that  $AG = DE$  and  $AH = DF$ .

$$\frac{AB}{DE} = \frac{AC}{DF} \quad (\text{given})$$

$$\frac{AB}{AG} = \frac{AC}{AH} \quad (\text{by construction})$$

$$\therefore GH \parallel BC \quad (\text{sides in prop.})$$

$$\therefore \hat{B} = \hat{AGH} \quad (\text{corresp. } \angle\text{s, } GH \parallel BC)$$

$$\text{and } \hat{C} = \hat{AHG} \quad (\text{corresp. } \angle\text{s, } GH \parallel BC)$$

$$\therefore \triangle AGH \equiv \triangle ABC \quad (\text{equiangular } \triangle\text{s})$$

$$\begin{aligned} \therefore \frac{GH}{BC} &= \frac{AG}{AB} \\ &= \frac{DE}{AB} \quad (AG = DE, \text{ by construction}) \\ &= \frac{EF}{BC} \quad (\text{given } \frac{AB}{DE} = \frac{BC}{EF}) \end{aligned}$$

$$\therefore GH = EF$$

$$\therefore \triangle AGH \equiv \triangle DEF \quad (\text{SSS})$$

$$\therefore \triangle ABC \text{ and } \triangle DEF \text{ are equiangular}$$

$$\therefore \triangle ABC \equiv \triangle DEF$$

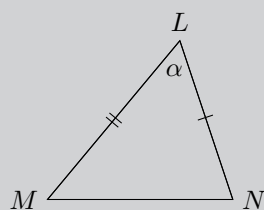
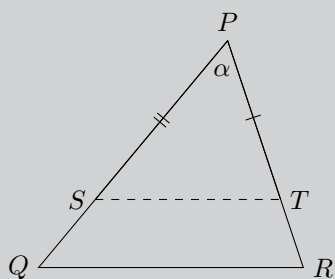
**Theorem:** Triangles with two sides in proportion and equal included angles, are similar (PROOF NOT FOR EXAMS)

### STATEMENT

If two sides of one triangle are in proportion to two sides of another triangle and the included angles are equal, then the two triangles are similar.

(Reason:  $\triangle$ s with 2 sides in prop. and equal incl.  $\angle$ s)





**Given:**

$\triangle PQR$  and  $\triangle LMN$  with  $\frac{LM}{PQ} = \frac{LN}{PR}$  and  $\hat{L} = \hat{P} = \alpha$

**Required to prove:**

$\hat{M} = \hat{Q}$  and  $\hat{N} = \hat{R}$

**PROOF**

**Construction:** draw  $ST$  such that  $PS = LM$  and  $PT = LN$ .

In  $\triangle PQR$  and  $\triangle LMN$ :

$$PS = LM \quad (\text{by construction})$$

$$\hat{P} = \hat{L} \quad (\text{given})$$

$$PT = LN \quad (\text{by construction})$$

$$\therefore \triangle PST \equiv \triangle LMN \quad (\text{SAS})$$

$$\therefore \hat{PST} = \hat{M}$$

$$\hat{PTS} = \hat{N}$$

$$\text{and } \frac{PS}{PQ} = \frac{PT}{PR} \quad (\text{by construction})$$

$$\therefore ST \parallel QR \quad (\text{sides in prop.})$$

$$\therefore \hat{Q} = \hat{PST} = \hat{M} \quad (\text{corresp. } \angle\text{s, } ST \parallel QR)$$

$$\text{and } \hat{R} = \hat{PTS} = \hat{N} \quad (\text{corresp. } \angle\text{s, } ST \parallel QR)$$

$$\therefore \triangle PQR \equiv \triangle LMN \quad (\text{equiangular } \triangle\text{s})$$

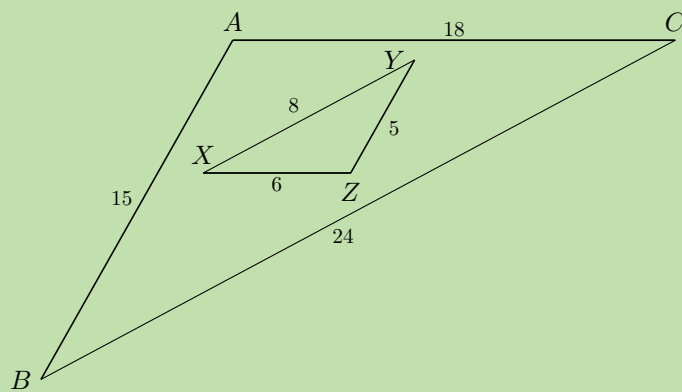
### Proving triangles with sides in proportion are similar:

To prove that the sides of two triangles are in the same proportion, we must show that all **three** pairs of corresponding sides are in the same proportion.

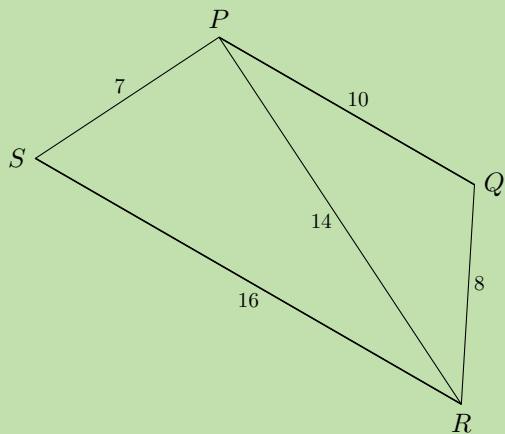
#### Worked example 6: Similarity of triangles

##### QUESTION

1. In the diagram below, is  $\triangle ABC \parallel \triangle ZYX$ ?  
Show calculations.



2. Consider the diagram given below. Determine whether or not  $\triangle PQR \parallel \triangle SPR$ .  
Show calculations.



(Diagrams are not drawn to scale)

### SOLUTION

#### Step 1: Investigate proportionality of $\triangle ABC$ and $\triangle ZYX$

For similar triangles, the order in which the triangles are labelled indicates which pairs of sides correspond.

In  $\triangle ABC$  and  $\triangle ZYX$  :

$$\frac{AB}{ZY} = \frac{15}{5} = \frac{3}{1}$$

$$\frac{AC}{ZX} = \frac{18}{6} = \frac{3}{1}$$

$$\frac{BC}{YX} = \frac{24}{8} = \frac{3}{1}$$

$$\therefore \triangle ABC \sim \triangle ZYX \quad (\text{SSS})$$

#### Step 2: Investigate proportionality of $\triangle PQR$ and $\triangle RPS$

In  $\triangle PQR$  and  $\triangle RPS$  :

$$\frac{PQ}{RP} = \frac{10}{14} = \frac{5}{7}$$

$$\frac{PR}{RS} = \frac{14}{16} = \frac{7}{8}$$

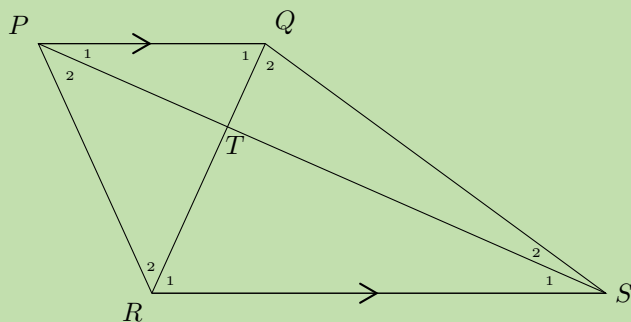
$$\frac{QR}{PS} = \frac{8}{7}$$

$$\therefore \triangle PQR \text{ is not similar to } \triangle RPS$$

### Worked example 7: Similarity of triangles

#### QUESTION

$PQSR$  is a trapezium, with  $PQ \parallel RS$ . Prove that  $PT \cdot RT = ST \cdot QT$ .

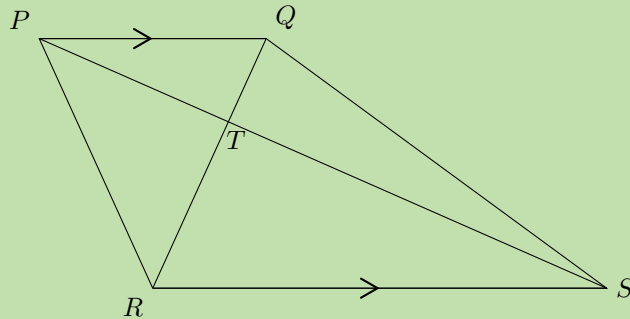


## SOLUTION

### Step 1: Identify triangles

We are required to prove that  $PT \cdot RT = ST \cdot QT$ , which we can also write as the ratio  $\frac{PT}{QT} = \frac{ST}{RT}$ .

To determine which two triangles we must consider, examine the lengths given in the ratio.



### Step 2: Prove triangles are equiangular

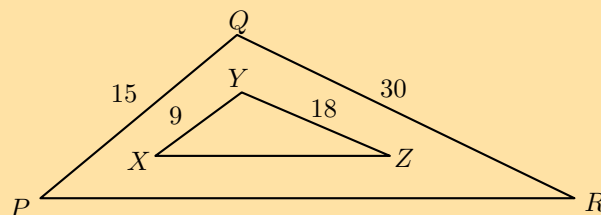
$$\begin{aligned} \text{In } \triangle PTQ \text{ and } \triangle STR: \quad \hat{P}_1 &= \hat{S}_1 && (\text{alt. } \angle\text{s, } PQ \parallel RS) \\ \hat{Q}_1 &= \hat{R}_1 && (\text{alt. } \angle\text{s, } PQ \parallel RS) \\ \therefore \triangle PTQ &\parallel\parallel \triangle STR && (\text{AAA}) \end{aligned}$$

### Step 3: Use proportionality

$$\begin{aligned} \frac{PT}{TQ} &= \frac{ST}{TR} && (\triangle PTQ \parallel\parallel \triangle STR) \\ \therefore PT \cdot TR &= ST \cdot TQ \end{aligned}$$

## Exercise 8 – 8: Similarity of triangles

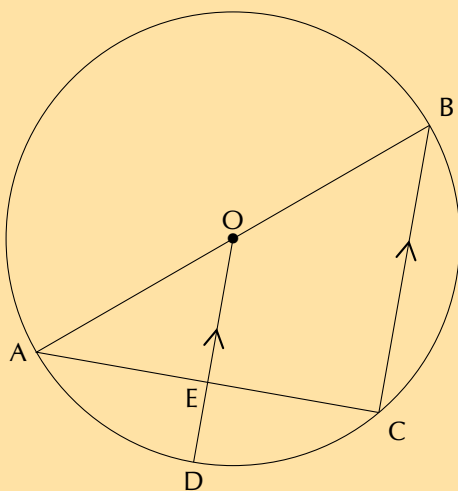
1. Consider the diagram given below.  $PR = 20$  units and  $XZ = 12$  units. Is  $\triangle XYZ \parallel\parallel \triangle PQR$ ? Give reasons.



2.  $AB$  is a diameter of the circle  $ABCD$ .  $OD$  is drawn parallel to  $BC$  and meets  $AC$  in  $E$ .

If the radius is 10 cm and  $AC = 16$  cm, calculate the length of  $ED$ .

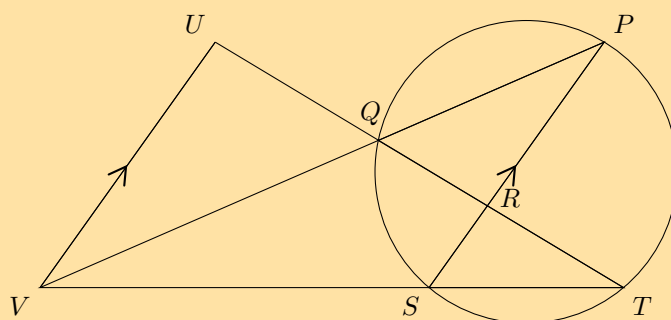
[NCS, Paper 3, November 2011]



3.  $P, Q, S$  and  $T$  are on the circumference of the circle.

$TS$  is produced to  $V$  so that  $SV = 2TS$ .

$TRQ$  is produced to  $U$  so that  $VU \parallel SRP$ .

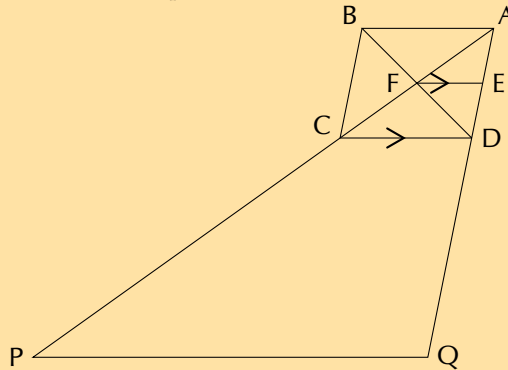


Prove, with reasons, that:

- $\frac{TR}{RU} = \frac{1}{3}$
- $\triangle TQV \parallel \triangle PSV$
- $QV \cdot PV = 6TS^2$
- $\triangle UQV \parallel \triangle RQP \parallel \triangle RST$

4.  $ABCD$  is a parallelogram with diagonals intersecting at  $F$ .  $FE$  is drawn parallel to  $CD$ .  $AC$  is produced to  $P$  such that  $PC = 2AC$  and  $AD$  is produced to  $Q$  such that  $DQ = 2AD$ .

[NCS, Paper 3, November 2011]



- Show that  $E$  is the midpoint of  $AD$ .
- Prove  $PQ \parallel FE$ .
- If  $PQ$  is 60 cm, calculate the length of  $FE$ .

5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 299Z    2. 29B2    3. 29B3    4. 29B4



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## 8.6 Pythagorean theorem

EMCJH

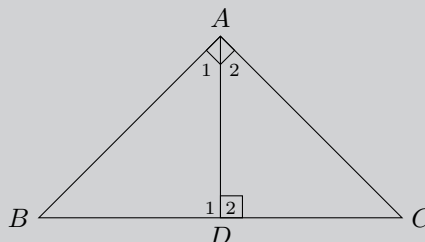
Many different methods of proving the theorem of Pythagoras have been formulated over the years. Similarity of triangles is one method that provides a neat proof of this important theorem.

### Theorem: Pythagorean theorem

#### STATEMENT

The square on the hypotenuse of a right-angled triangle is equal to the sum of the squares on the other two sides.

(Reason: Pythagoras or right-angled  $\triangle$ s)



**Given:**

$\triangle ABC$  with  $\hat{A} = 90^\circ$

**Required to prove:**

$$BC^2 = AB^2 + AC^2$$

## PROOF

---

**Construction:** draw  $AD \perp BC$ .

$$\hat{C} + \hat{A}_2 = 90^\circ \quad (\angle\text{s of } \triangle CAD)$$

$$\hat{A}_1 + \hat{A}_2 = 90^\circ \quad (\text{given})$$

$$\therefore \hat{A}_1 = \hat{C}$$

$$\hat{A}_1 + \hat{B} = 90^\circ \quad (\angle\text{s of } \triangle ABD)$$

$$\therefore \hat{B} = \hat{A}_2$$

$$\hat{D}_1 = \hat{D}_2 = \hat{A} = 90^\circ \quad (\text{construction})$$

$$\therefore \triangle ABD \parallel \triangle CBA \parallel \triangle CAD \quad (\text{AAA})$$

$$\therefore \frac{AB}{BC} = \frac{BD}{AB} \quad (\triangle ABD \parallel \triangle CBA)$$

$$AB^2 = BD \times BC$$

$$\text{Similarly } \frac{AC}{CB} = \frac{DC}{AC} \quad (\triangle ABD \parallel \triangle CBA)$$

$$AC^2 = CB \times DC$$

$$\begin{aligned} \therefore AC^2 + AB^2 &= (BD \times BC) + (CB \times DC) \\ &= BC(BD + DC) \\ &= BC(BC) \\ &= BC^2 \end{aligned}$$

$$\therefore BC^2 = AC^2 + AB^2$$

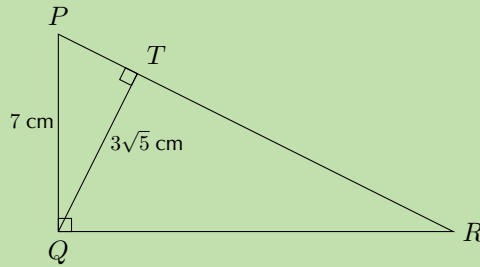
### Converse: theorem of Pythagoras

If the square of one side of a triangle is equal to the sum of the squares of the other two sides of the triangle, then the angle included by these two sides is a right angle.

### Worked example 8: Theorem of Pythagoras

#### QUESTION

In  $\triangle PQR$ ,  $\hat{P}QR = 90^\circ$  and  $QT \perp PR$ . If  $PQ = 7$  cm and  $QT = 3\sqrt{5}$  cm, determine  $PR$  and  $QR$  (correct to the nearest integer).



#### SOLUTION

**Step 1: Use the theorem of Pythagoras to determine PT**

$$\begin{aligned}\text{In } \triangle PTQ, \quad PT^2 &= PQ^2 - QT^2 && \text{(Pythagoras)} \\ &= 7^2 - (3\sqrt{5})^2 \\ &= 49 - 45 \\ \therefore PT &= \sqrt{4} \\ &= 2 \text{ cm}\end{aligned}$$

**Step 2: Use proportionality to determine PR and QR**

$$\begin{aligned}\hat{P}QR &= 90^\circ && \text{(given)} \\ QT &\perp PR && \text{(given)} \\ \therefore \triangle PQT &\parallel \triangle QRT \parallel \triangle PRQ && \text{(right-angled } \triangle\text{s)} \\ \therefore \frac{QT}{TR} &= \frac{PT}{QT} && (\triangle QRT \parallel \triangle PQT) \\ \therefore QT^2 &= TR \cdot PT\end{aligned}$$

$$\begin{aligned}(3\sqrt{5})^2 &= TR \cdot 2 \\ \frac{45}{2} &= TR\end{aligned}$$

$$\begin{aligned}\text{And } PR &= PT + TR \\ &= \frac{45}{2} + 2 \\ &= 25 \text{ cm} && \text{(to nearest integer)}\end{aligned}$$

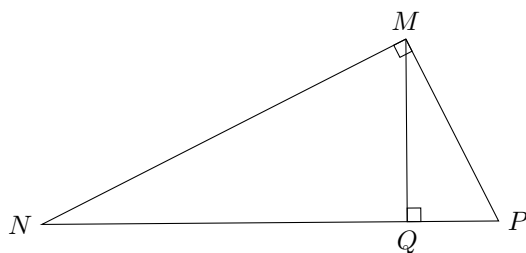
$$\begin{aligned}\text{In } \triangle PQR, \quad QR^2 &= PR^2 - PQ^2 && \text{(Pythagoras)} \\ &= 25^2 - 7^2 \\ \therefore QR &= \sqrt{576} \\ &= 24 \text{ cm}\end{aligned}$$

**Step 3: Write the final answer**

$PR = 25$  cm and  $QR = 24$  cm



For any right-angled  $\triangle MNP$ , if  $MQ$  is drawn perpendicular to  $NP$ , then:



$$\triangle MNQ \parallel \triangle PMQ \implies MQ^2 = NQ \cdot PQ$$

$$\triangle MPQ \parallel \triangle NPM \implies MP^2 = QP \cdot NP$$

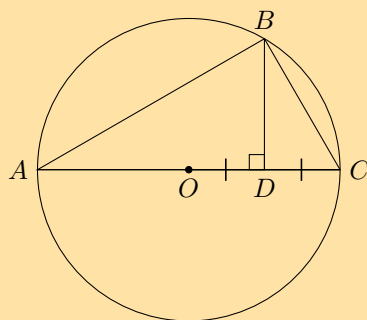
$$\triangle MNQ \parallel \triangle PNM \implies MN^2 = PN \cdot QN$$

### Exercise 8 – 9: Theorem of Pythagoras

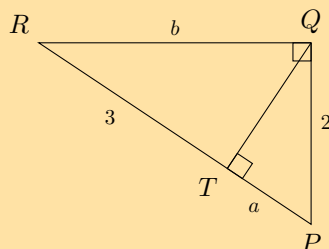
1.  $B$  is a point on circle with centre  $O$ .  $BD \perp AC$  and  $D$  is the midpoint of radius  $OC$ .

If the diameter of the circle is 24 cm, find  $BD$ .

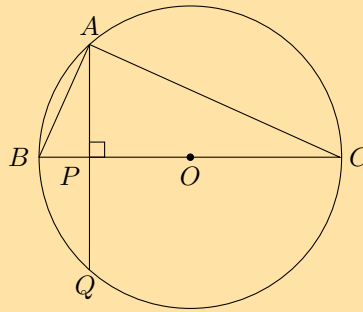
Leave answer in simplified surd form.



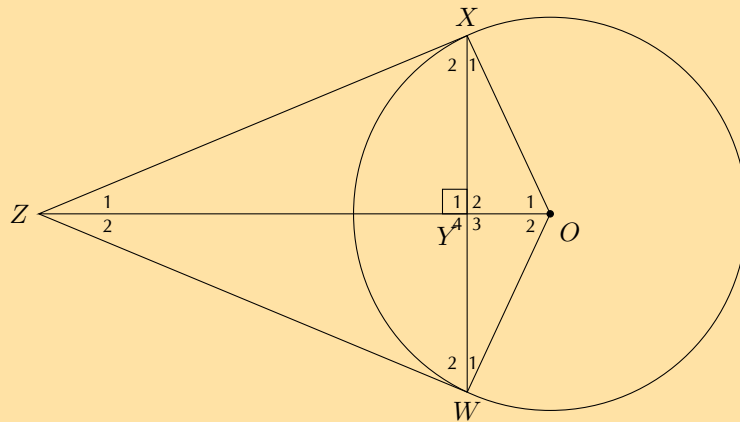
2. In  $\triangle PQR$ ,  $RQ \perp QP$  and  $QT \perp RP$ .  $PQ = 2$  units,  $QR = b$  units,  $RT = 3$  units and  $TP = a$  units. Determine  $a$  and  $b$ , giving reasons.



3. Chord  $AQ$  of circle with centre  $O$  cuts  $BC$  at right angles at point  $P$ .



- Why is  $\triangle ABP \parallel \triangle CBA$ ?
  - If  $AB = \sqrt{6}$  units and  $PO = 2$  units, calculate the radius of the circle.
4. In the diagram below,  $XZ$  and  $WZ$  are tangents to the circle with centre  $O$  and  $\angle XYZ = 90^\circ$ .



- Show that  $XY^2 = OY \cdot YZ$ .
  - Prove that  $\frac{OY}{YZ} = \frac{OW^2}{WZ^2}$ .
5. More questions. Sign in at Everything Maths online and click 'Practise Maths'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 29B5    2. 29B6    3. 29B7    4. 29B8



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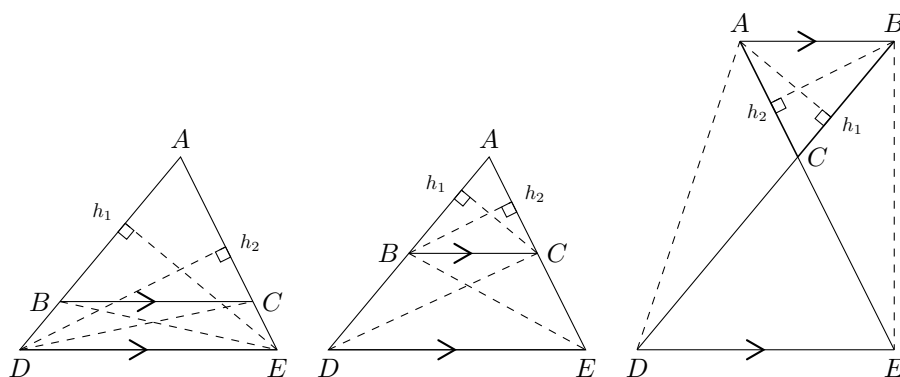


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- A ratio describes the relationship between two quantities which have the same units.

$$x : y \quad \text{or} \quad \frac{x}{y} \quad \text{or} \quad x \text{ to } y$$

- If two or more ratios are equal to each other  $\left(\frac{m}{n} = \frac{p}{q}\right)$ , then  $m$  and  $n$  are in the same proportion as  $p$  and  $q$ .
- A polygon is a plane, closed shape consisting of three or more line segments.
- Triangles with equal heights have areas which are proportional to their bases.
- Triangles with equal bases and between the same parallel lines are equal in area.
- Triangles on the same side of the same base and equal in area lie between parallel lines.
- A line drawn parallel to one side of a triangle divides the other two sides of the triangle in the same proportion.



$$\frac{AD}{DB} = \frac{AE}{EC}$$

and

$$\frac{AB}{AD} = \frac{AC}{AE}$$

and

$$\frac{CD}{BD} = \frac{CE}{AE}$$

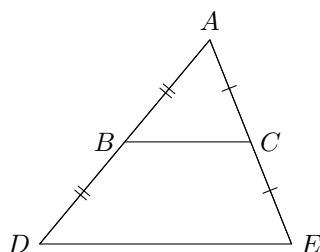
- **Converse: proportion theorem**

If a line divides two sides of a triangle in the same proportion, then the line is parallel to the third side.

- **Special case: the mid-point theorem**

The line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.

If  $AB = BD$  and  $AC = CE$ , then  $BC \parallel DE$  and  $BC = \frac{1}{2}DE$ .

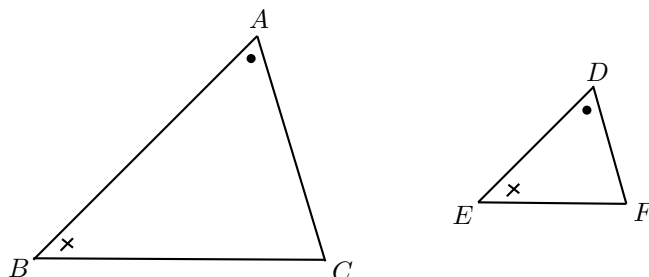


- **Converse: the mid-point theorem**

The line drawn from the mid-point of one side of a triangle parallel to another side, bisects the third side of the triangle.

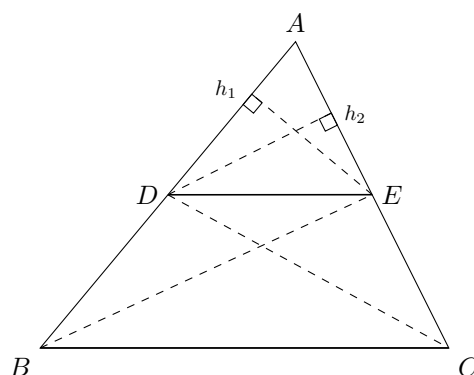
If  $AB = BD$  and  $BC \parallel DE$ , then  $AC = CE$ .

- Polygons are similar if they are the same shape but differ in size. One polygon is an enlargement of the other.
- Two polygons with the same number of sides are similar when:
  1. All pairs of corresponding angles are equal, **and**
  2. All pairs of corresponding sides are in the same proportion.
- If two triangles are equiangular, then the triangles similar.



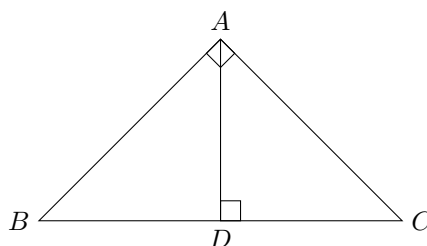
$$\triangle ABC \parallel \triangle DEF$$

- Triangles with sides in proportion are equiangular and therefore similar.



$$\triangle ADE \parallel \triangle ABC$$

- The square on the hypotenuse of a right-angled triangle is equal to the sum of the squares on the other two sides.



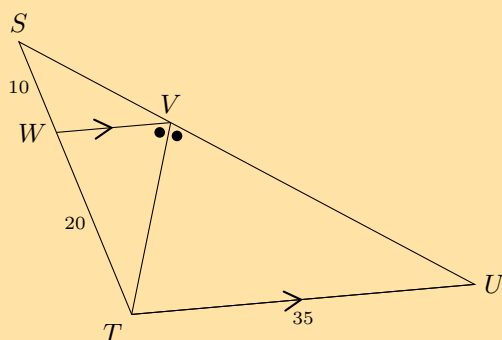
$$BC^2 = AC^2 + AB^2$$

- **Converse: theorem of Pythagoras**

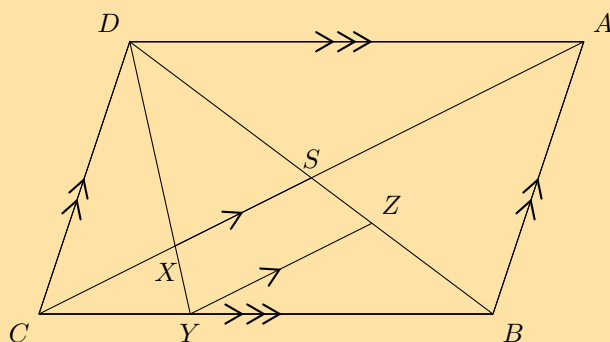
If the square of one side of a triangle is equal to the sum of the squares of the other two sides of the triangle, then the angle included by these two sides is a right angle.

# Exercise 8 – 10: End of chapter exercises

1. Calculate  $SV$

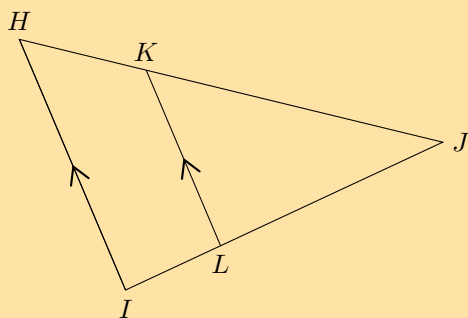


2.  $\frac{CB}{YB} = \frac{3}{2}$ . Find  $\frac{DS}{SZ}$ .

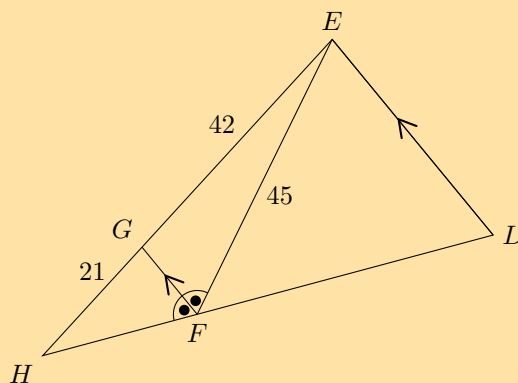


3. Using the following figure and lengths, find  $IJ$  and  $KJ$  (correct to one decimal place).

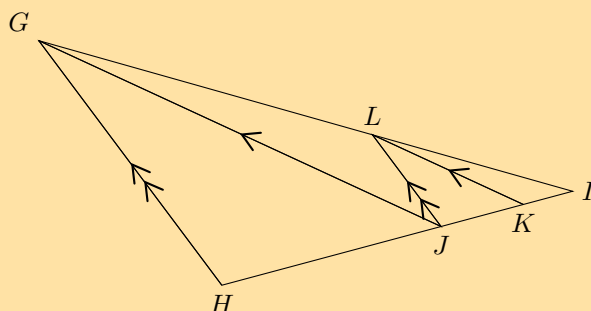
$HI = 20$  m,  $KL = 14$  m,  $JL = 18$  m and  $HJ = 32$  m.



4. Find  $FH$  in the following figure.

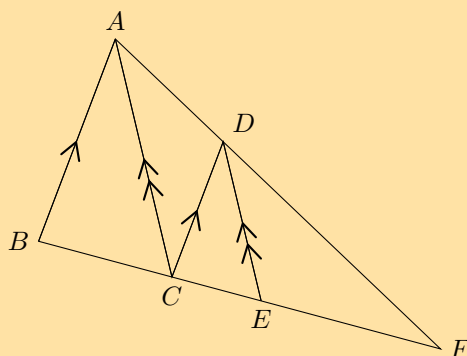


5. In  $\triangle GHI$ ,  $GH \parallel LJ$ ,  $GJ \parallel LK$  and  $\frac{JK}{KI} = \frac{5}{3}$ . Determine  $\frac{HJ}{KI}$ .

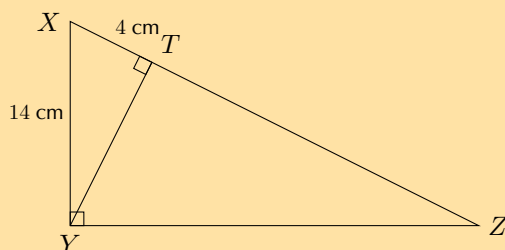


6.  $BF = 25$  m,  $AB = 13$  m,  $AD = 9$  m,  $DF = 18$  m.

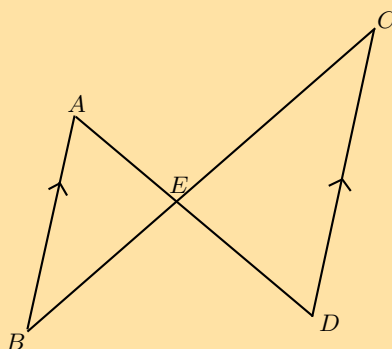
Calculate the lengths of  $BC$ ,  $CF$ ,  $CD$ ,  $CE$  and  $EF$ , and find the ratio  $\frac{DE}{AC}$ .



7. In  $\triangle XYZ$ ,  $\hat{XYZ} = 90^\circ$  and  $YT \perp XZ$ . If  $XY = 14$  cm and  $XT = 4$  cm, determine  $XZ$  and  $YZ$  (correct to two decimal places).

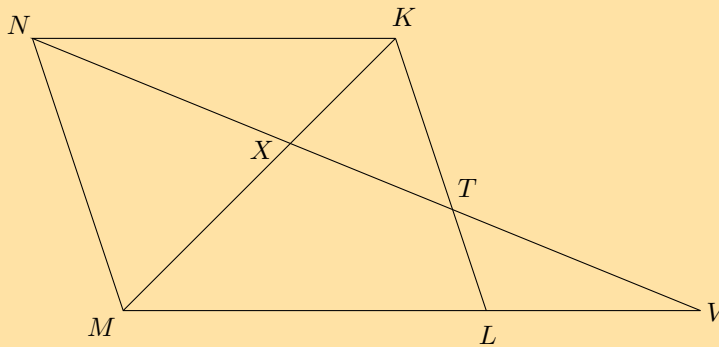


8. Given the following figure with the following lengths, find  $AE$ ,  $EC$  and  $BE$ .  
 $BC = 15$  cm,  $AB = 4$  cm,  $CD = 18$  cm, and  $ED = 9$  cm.



9.  $NKLM$  is a parallelogram with  $T$  on  $KL$ .

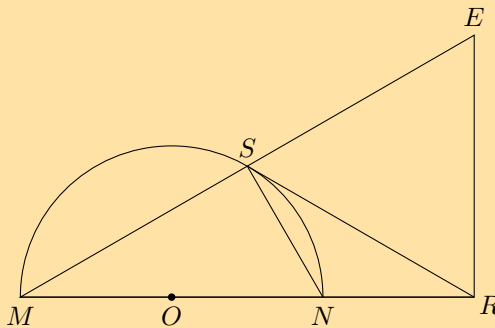
$NT$  produced meets  $ML$  produced at  $V$ .  $NT$  intercepts  $MK$  at  $X$ .



- Prove that  $\frac{XT}{NX} = \frac{XK}{MX}$ .
- Prove  $\triangle VXM \parallel \triangle NXK$ .
- If  $XT = 3$  cm and  $TV = 4$  cm, calculate  $NX$ .

10.  $MN$  is a diameter of circle  $O$ .  $MN$  is produced to  $R$  so that  $MN = 2NR$ .

$RS$  is a tangent to the circle and  $ER \perp MR$ .  $MS$  produced meets  $RE$  at  $E$ .

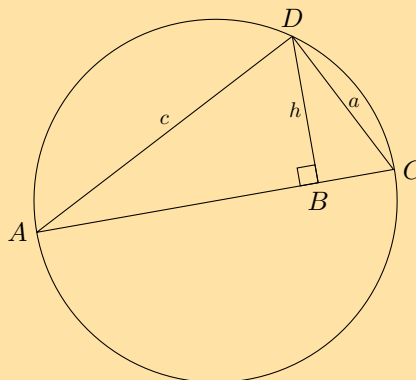


Prove that:

- $SNRE$  is a cyclic quadrilateral
- $RS = RE$
- $\triangle MSN \parallel \triangle MRE$
- $\triangle RSN \parallel \triangle RMS$
- $RE^2 = RN \cdot RM$

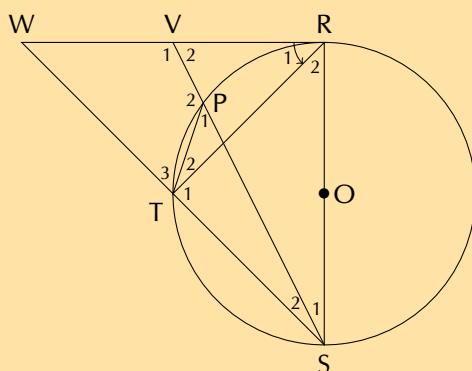
11.  $AC$  is a diameter of circle  $ADC$ .  $DB \perp AC$ .

$AC = d$ ,  $AD = c$ ,  $DC = a$  and  $DB = h$ .



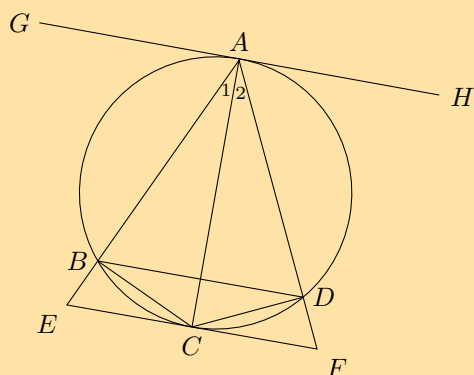
- Prove that  $h = \frac{ac}{d}$ .
- Hence, deduce that  $\frac{1}{h^2} = \frac{1}{a^2} + \frac{1}{c^2}$ .

12.  $RS$  is a diameter of the circle with centre  $O$ . Chord  $ST$  is produced to  $W$ . Chord  $SP$  produced meets the tangent  $RW$  at  $V$ .  $\hat{R}_1 = 50^\circ$ . [NCS, Paper 3, November 2011]



- Calculate the size of  $\hat{WRS}$ .
- Find  $\hat{W}$ .
- Determine the size of  $\hat{P}_1$ .
- Prove that  $\hat{V}_1 = \hat{PTS}$ .

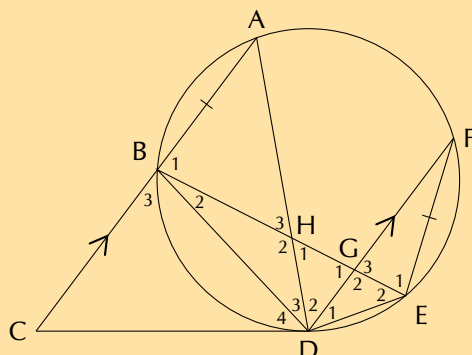
13.  $ABCD$  is a cyclic quadrilateral and  $BC = CD$ .  
 $ECF$  is a tangent to the circle at  $C$ .  $ABE$  and  $ADF$  are straight lines.



Prove:

- $AC$  bisects  $\hat{EAF}$
- $BD \parallel EF$
- $\triangle ADC \parallel \triangle CBE$
- $DC^2 = AD \cdot BE$

14.  $CD$  is a tangent to circle  $ABDEF$  at  $D$ . Chord  $AB$  is produced to  $C$ . Chord  $BE$  cuts chord  $AD$  in  $H$  and chord  $FD$  in  $G$ .  $AC \parallel FD$  and  $E = AB$ . Let  $D_4 = x$  and  $D_1 = y$ . [NCS, Paper 3, November 2011]



- Determine THREE other angles that are each equal to  $x$ .
- Prove that  $\triangle BHD \parallel \triangle FED$ .
- Hence, or otherwise, prove that  $AB \cdot BD = FD \cdot BH$ .

15. More questions. Sign in at Everything Maths online and click 'Practise Maths'.  
 Check answers online with the exercise code below or click on 'show me the answer'.

- |          |          |         |          |          |          |
|----------|----------|---------|----------|----------|----------|
| 1. 29B9  | 2. 29BB  | 3. 29BC | 4. 29BD  | 5. 29BF  | 6. 29BG  |
| 7. 29BH  | 8. 29BJ  | 9. 29BK | 10. 29BM | 11. 29BN | 12. 29BP |
| 13. 29BQ | 14. 29BR |         |          |          |          |



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